

Tensor saturation for even spin groups

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Abstract

We prove the saturation conjecture for $\text{Spin}(2n)$, $n \geq 2$. If three dominant integral weights sum to an element of the root lattice, then the existence of a nonzero tensor invariant after a positive integral dilation implies the existence of one at the original weights.

1 Introduction

Let G be a simply connected complex semisimple group, and write $V(\lambda)$ for its irreducible representation of dominant integral highest weight λ . The saturation problem asks whether occurrence of a tensor invariant at dilated highest weights forces occurrence at the original weights. A necessary congruence is that the sum of the original weights belong to the root lattice: otherwise the center acts nontrivially on the tensor product. Write Q for the root lattice. The question is therefore whether the semigroup of triples with a tensor invariant contains every triple of dominant integral weights in its rational cone whose sum belongs to Q .

This arithmetic question is closely connected with the geometry of eigenvalues. For compact groups, the corresponding real cone describes which three coadjoint orbits contain representatives with sum zero. Klyachko connected the Hermitian eigenvalue problem with tensor occurrence after dilation [23]. The relation between moment polytopes and occurrence after integral dilation is part of the general branching framework of Berenstein and Sjamaar [5, Theorems 3.2.1–3.2.2]. Saturation asks whether this real feasibility, together with the central congruence, already detects an invariant at the original weights.

Knutson and Tao’s honeycomb theorem answered this question in type A [24]. Derksen and Weyman gave a quiver proof through saturation of semi-invariant weights [7], and Belkale gave a geometric proof [3]. Kapovich and Millson formulated the analogous root-lattice assertion for simply laced groups [20, Conjecture 1.4(1)]; see also [17, Conjecture 1.5]. We prove its type- D case.

Theorem 1.1 (Type- D saturation). *Let $G = \text{Spin}(2n)$, $n \geq 2$, and let λ, μ, ν be dominant integral weights such that $\lambda + \mu + \nu \in Q(D_n)$. Then, for every positive integer N ,*

$$(V(\lambda) \otimes V(\mu) \otimes V(\nu))^G \neq 0 \iff (V(N\lambda) \otimes V(N\mu) \otimes V(N\nu))^G \neq 0.$$

The theorem includes the half-integral spin weights. We use the standard convention $D_2 = A_1 \times A_1$; D_3 is isomorphic to A_3 . If $n = 1$ is included, G is a torus and the assertion is just the weight-sum equation. The stretching parameter throughout is a positive integer.

Kapovich, Kumar, and Millson proved saturation for $\text{Spin}(8)$ [17, Theorem 5.3], and Kiers proved it for $\text{Spin}(10)$ and $\text{Spin}(12)$ [22, Theorem 1.3]. These results include the spin representations

and establish the first nontrivial ranks beyond the type- A identifications. General saturation-factor results give another important approach. The building and path methods of Kapovich and Millson give a factor of four in type D [19, Theorem 1.1]. Belkale and Kumar obtained factor two for symplectic and odd special orthogonal groups [4]. Sam’s symmetric-quiver methods give factor two uniformly for special orthogonal and symplectic groups and, by descent of doubled weights, factor four for spin groups [33, Theorem 1.1 and Corollary 1.2]. These factors do not give the scale-one conclusion under the root-lattice hypothesis. Hong and Shen subsequently obtained factor two for odd spin groups under the root-lattice congruence [15, Corollary 1.3]. Recent progress also includes Min’s proof of saturation for Newell–Littlewood numbers for triples of partitions of even total size [30, Theorem 1.2]. These numbers describe stable ordinary-weight tensor multiplicities for the classical groups; that range excludes general spin weights and the full unstable range covered here.

Our proof uses paths, but separates their geometric and arithmetic roles from the construction of representations. Littelmann’s path model encodes tensor products through root operators and integral saturated chains [26, 27]. Kapovich and Millson relate such paths to geodesics in buildings [19], while Kapovich, Leeb, and Millson connect weighted configurations with polygons in symmetric spaces [18]. We use this geometric viewpoint to obtain real paths, then solve the type- D lattice problem by deforming their switches. The resulting integral schedules have a weaker contract than Lakshmibai–Seshadri paths. A separate construction with exterior and spin columns supplies tensor occurrence.

We describe the main constructions. Put $\xi = \nu^*$, so that the desired invariant is equivalent to an intertwiner $V(\lambda) \otimes V(\mu) \longrightarrow V(\xi)$. A *shape* is a polygonal path from 0 to μ with dominant increments. A *schedule* starts at λ , ends in the Weyl orbit of ξ , and follows Weyl translates of these increments, changing direction by suitably oriented root reflections. At a reflection, the difference between the outgoing and incoming Weyl translates of the shape value is a scalar multiple of the reflecting root. This scalar is the switch coefficient; a schedule is integral when every such coefficient is an integer.

The first construction produces a schedule whose switch roots form a real basis. An area maximum and a torus-orbit polytope argument also make its switches limits of Bruhat covers. A real root basis need not generate the full root lattice, and this is where integrality can fail. In type D , rank reduction leaves only one obstruction: the signed root graph has two unbalanced components, each with an odd coordinate sum in its assigned label difference. Joining the components introduces one relation among the roots. Motion along half its primitive integer generator reaches integral switch coefficients without changing the endpoint.

Reaching those coefficients arithmetically does not yet give a path: the switch times and signs must remain admissible throughout the motion. Rank-two reorderings continue any locally feasible join through simultaneous switches. A strict spectral multiplicity count, compared with the evolution of signed coordinate comparisons, forces such a join even at exact ties. Together these arguments give an integral schedule whenever the moving weight has at most one zero coordinate.

A *fundamental column* is a straight path segment whose increment is a fundamental weight. The second construction converts integral schedules for a sequence of such columns into tensor intertwiners. Minuscule columns have no effective interior switches. An ordinary exterior column can change only at its midpoint. A common-wall invariant covector, built from coordinate determinants and skew pairing forms, has a nonzero coefficient at a suitable flag. A flag-composition argument then places the product of these coefficients on the Cartan summand for μ . This supplies the tensor occurrence without strengthening the switch condition to a saturated path chain.

Finally, a preprojective-module construction removes a common terminal zero block. Its main linear-algebra ingredient is a filtered intersection estimate for the socle of a nilpotent operator.

Doubling Jordan rows with opposite alternating colors allows the top and socle bounds to hold simultaneously at new orthogonal leaves. The doubling is used only to obtain a smaller-rank real moment solution; the integral schedule is then constructed at the original scale and padded back to the original columns. These three constructions separate the real geometry, the root-lattice obstruction, and the recovery of tensor coefficients.

Section 2 develops the real schedules and their basis property. Sections 3 and 4 prove their integrality. Section 5 constructs the column intertwiners, and Section 6 proves the zero-tail reduction and completes Theorem 1.1.

2 Real moments and schedules

We establish both directions between compact-orbit addition and schedules. From a real moment solution we construct a schedule whose switch roots form a basis. Conversely, a schedule gives a real moment solution; this implication will be used for the block decompositions in the next section. Neither statement assumes a root-lattice condition. The construction works in type D and in coordinate permutation systems, with orthogonal central coordinates carried along by ordinary addition.

2.1 Conventions and the path problem

All roots have squared length two. In type D_l we use

$$\Phi = \{\pm e_i \pm e_j : i \neq j\}, \quad \mathcal{C} = \{x_1 \geq \cdots \geq x_{l-1} \geq |x_l|\}.$$

The root lattice consists of the integral vectors of even coordinate sum, and the weight lattice consists of the vectors whose coordinates are either all integral or all half-odd-integral. We include $D_2 = A_1 \times A_1$. In a permutation system the roots are $e_i - e_j$; the common-coordinate direction is central. More generally, whenever a root subsystem has an orthogonal central complement, we apply the semisimple arguments after subtracting that complement and retain its ordinary vector-addition equation. Its coordinates need not be integrally normalized.

We use addition labels λ, μ, ξ , where $\xi = \nu^*$ is the dominant representative of $-\nu$. Thus the root-lattice condition in Theorem 1.1 is equivalent to $\lambda + \mu - \xi \in Q$. Let W be the Weyl group. A *shape* ending at μ is a continuous polygonal path $\eta : [0, 1] \rightarrow \mathbb{R}^l$ with $\eta(0) = 0$, $\eta(1) = \mu$, and every increment in $\mathcal{C}$. It is *regular* if all its slopes lie in the open chamber.

Definition 2.1 (Schedule). A schedule for (λ, η, ξ) starts at λ , ends at a point of $W\xi$, and, between finitely many ordered switch times, satisfies $y' = w\eta'$ for a fixed $w \in W$. At a switch at time t , the frame changes from w to $s_\beta w$, where β is a signed root satisfying

$$G = y(t) \cdot \beta \geq 0, \quad -\beta \cdot w\mu \geq 0. \quad (2.1)$$

The number G is the position gap, and $-\beta \cdot w\mu$ is the total moving-weight pairing. Several switches may occur at the same time, with their order part of the data. The switch coefficient is $m = -\beta \cdot w\eta(t)$. The schedule is *integral* if every switch coefficient is an integer.

If the current frame is w and the preceding switches have coefficients m_i and position roots β_i , continuity gives

$$y(t) = \lambda + w\eta(t) - \sum_{i \text{ preceding } t} m_i \beta_i. \quad (2.2)$$

The formula uses the chosen ordered prefix at a multiple switch time. Indeed $s_\beta w\eta = w\eta + m\beta$ at a switch, exactly canceling the newly added term in the sum.

There are three useful simplifications. A switch with $-\beta \cdot w\mu = 0$ changes no shape increment: the root $w^{-1}\beta$ has one sign on the dominant chamber, so its zero total pairing forces zero pairing with every increment. Such a switch can be deleted, changing later direction indices on the right by a stabilizer of the entire shape if necessary. Switches at 0 or 1 can be absorbed into the initial frame or the endpoint direction. Finally, if $G = 0$, reflect the whole suffix, including its switch roots, in that wall and delete the switch. The path remains continuous and its endpoint remains in $W\xi$. These operations also preserve integrality when the original schedule is integral.

Let $\mathcal{G}$ be the simply connected complex semisimple group and K a compact form; the matrix models below describe their compact orbit quotients. We identify real Cartan vectors with Hermitian generators in $i\mathfrak{k}$. A *real moment solution* for (λ, μ, ξ) is a zero-sum triple in the compact orbits of $\lambda, \mu, -\xi$. In type D , these are i times real skew-symmetric matrices. Their eigenvalues are the positive and negative signed coordinates; the compact orbits are special orthogonal orbits, so their orientation information is retained. In permutation systems we use Hermitian matrices. The central equation is imposed separately. Compactness of K shows that moment solvability is closed in the three labels.

Write $B = TU^+$ and $B^- = TU^-$, with B fixing highest lines. We index the torus-fixed points of $\mathcal{G}/B$ by W , and use the Bruhat convention

$$u \geq v \iff vB \in \overline{BuB/B}.$$

Downward reflection steps have the form $u \mapsto s_\beta u$ with $\beta > 0$ and $u^{-1}\beta < 0$. A Bruhat cover is a strict comparison with no intervening Weyl element. Inversion preserves Bruhat order. For a regular dominant ρ , the flow $a(t) = \exp(t\rho)$ has past set U^+p and future set U^-p at a fixed flag p .

The principal construction has the following precise output. Its cover condition is read in the chamber coordinates obtained by folding the path into $\mathcal{C}$.

Proposition 2.2 (Basic real schedule). *Let the root system be of type D or a permutation system, possibly with an orthogonal central space. If (λ, μ, ξ) has a real moment solution, then every shape η ending at μ admits a schedule whose switch roots form a real basis of the semisimple space. Weak signs, endpoint times, and coincident times are allowed. More precisely, it is a limit of schedules for nearby regular data whose switches are covers in their local chamber frames. Simplifying the limiting schedule may leave fewer than rank-many switches.*

We prove the proposition after constructing decreasing paths and maximizing their area. First we develop the coefficient tools needed for the converse from schedules to moments.

2.2 Coefficients and flags

The two tools below also combine the fundamental columns in Section 5. The classical orthogonal representation models are recalled in [10, Lectures 19–20]. We give the coefficient and composition arguments explicitly because the second must control restriction to a Cartan summand.

For an integral dominant h , write v_h for a highest vector of $V(h)$, and write f_i for the negative simple generators. A flag vector of weight h at a flag b means a nonzero vector on the translate of the line $\mathbb{C}v_h$ associated with b .

The coefficient description below is the dual form of the Parthasarathy–Ranga Rao–Varadarajan tensor-multiplicity formula [32]; see also [31, Theorems 1.1–1.2]. For the classical coefficient criterion, see Kumar [25, Theorem 3.7], where it is attributed to Kostant. The already-dominant

extremal consequence appears in [25, Corollary 3.8]. We include the argument to fix the highest-coordinate convention.

Lemma 2.3 (Highest-to-highest coefficients). *Let h, h', θ be dominant integral weights. The linear forms obtained by taking the highest-output coordinate of*

$$T(v_h \otimes \cdot), \quad T : V(h) \otimes V(\theta) \longrightarrow V(h'),$$

are exactly the forms supported at weight $h' - h$ that annihilate

$$\sum_i f_i^{h \cdot \alpha_i + 1} V(\theta). \quad (2.3)$$

In particular, if $h' = h + u\theta$ is dominant, the extremal-coordinate functional at $u\theta$ gives such an intertwiner and is nonzero on the flag vector at u .

Proof. Take coinvariants for the diagonal negative nilpotent algebra on the tensor product. At weight h' this quotient records exactly the highest coordinate of each copy of $V(h')$. Move negative generators from the first tensor factor to the second by the antipode. The highest-generator presentation

$$V(h) = U(\mathfrak{n}^-) / \sum_i U(\mathfrak{n}^-) f_i^{h \cdot \alpha_i + 1}$$

then gives precisely the quotient by (2.3): the antipode reverses products, so a relation $X f_i^{h \cdot \alpha_i + 1}$ transfers into the image of the indicated power on an arbitrary second vector. This proves both necessity and sufficiency after dualizing the weight quotient.

The presentation itself follows by imposing the simple singular-vector relations on the Verma module. Each simple $\mathfrak{sl}_2$ then acts integrably: local nilpotence extends from the generating vector using the adjoint action. The weights are therefore Weyl-invariant and bounded in the highest-weight order; together with PBW this makes the quotient finite-dimensional. Complete reducibility identifies it with the irreducible highest-weight module.

For an extremal weight $u\theta$, the upward simple-string length is $\max(0, -u\theta \cdot \alpha_i)$. It is at most $h \cdot \alpha_i$ because $h + u\theta$ is dominant. Thus its coordinate functional annihilates every image in (2.3). $\square$

Products of suitably ordered flag coordinates are central to standard monomial theory; see, for example, [28, Theorem 4]. The following form works directly with the coefficients just constructed. Its ordering hypothesis is what ensures that a nonzero product survives on the Cartan summand.

Lemma 2.4 (Flag composition). *Suppose that $h_0, \dots, h_s$ and $\theta_1, \dots, \theta_s$ are dominant integral weights and that there are intertwiners*

$$T_j : V(h_{j-1}) \otimes V(\theta_j) \longrightarrow V(h_j).$$

Assume their highest-to-highest coefficients are nonzero on flag vectors at flags b_j . If the past and future limits of b_j under $a(t)$ are u_j, v_j , and $v_j \geq u_{j+1}$, then the iterated composition is nonzero on

$$V(h_0) \otimes V(\theta_1 + \dots + \theta_s).$$

Here the second factor is the Cartan summand in the tensor product of the $V(\theta_j)$.

Proof. The empty composition is the identity; assume $s \geq 1$. We first explain why one positive-flow orbit can approximate the waypoints $b_1, \dots, b_s$ in their prescribed order, with successive time gaps tending to infinity. When $v \geq u$, there is a bridge in $U^+v \cap U^-u$. Indeed U^+v approaches u , and a neighborhood of u is covered by small U^+U^- translates. Removing the small positive factor from a point of U^+v leaves a point in that intersection.

It remains to glue at a shared fixed flag p . Suppose the preceding waypoint approximation lies at $n^-p \in U^-p$ and the next desired point is $n^+p \in U^+p$. Replace the former by

$$n^-a(-t)n^+a(t)p. \quad (2.4)$$

As $t \rightarrow \infty$ this tends to n^-p , whereas after flowing for time t it tends to n^+p . The remaining negative factor after that flow preserves the future cell containing n^+p . Insert the bridges and repeat this construction, choosing each new time gap sufficiently large to preserve every earlier approximation.

For each such approximation, choose flag vectors $z_j \in V(\theta_j)$ at its common starting flag z , and times $t_1 < \dots < t_s$. Let C denote the iterated composition of the T_j , and set

$$F_j(x) = T_j(x \otimes a(t_j)z_j), \quad C_z(x) = C(x \otimes z_1 \otimes \dots \otimes z_s).$$

Equivariance gives the exact identity

$$F_s a(t_s - t_{s-1}) F_{s-1} \dots a(t_2 - t_1) F_1 = a(t_s) C_z a(-t_1).$$

Indeed, writing each F_j as $a(t_j)T_j(a(-t_j) \cdot \otimes z_j)$ cancels all the intermediate frame factors. For $s = 1$ the identity is just this equivariance formula.

Normalize each translated flag input so that it approaches the prescribed vector at b_j . Normalize each intervening $a(t_{j+1} - t_j)$ by its highest eigenvalue on $V(h_j)$; as the time gaps tend to infinity, it approaches projection onto the highest line. The highest-input, highest-output coordinate of the left side therefore tends to the product of the nonzero waypoint coefficients. It is nonzero for some finite approximation. On the right side the outer frame factors act by nonzero scalars on those coordinates, so C_z is nonzero. The z_j lie at one common flag, and hence their tensor belongs to the Cartan summand. This proves the assertion. $\square$

The next compact maximization argument is a direct instance of the norm methods of Kempf and Ness [21]. We need only the invariant-to-moment implication, whose proof follows.

Lemma 2.5 (Invariants give moments). *A nonzero invariant in $V(\lambda) \otimes V(\mu) \otimes V(\xi^*)$ gives a real moment solution for (λ, μ, ξ) . The same conclusion follows from an invariant for simultaneous positive integer multiples of the labels.*

Proof. Take an invariant functional and maximize its absolute evaluation on triples of unit flag vectors, using compact-invariant Hermitian norms. The maximum is attained and is nonzero, since the flag-vector orbits span their representations. Vary all three vectors simultaneously by a Hermitian generator H and renormalize. Invariance makes the unnormalized numerator constant; differentiation of the three norms therefore gives zero sum of the expectations of H . Since this holds for all H , the three moments sum to zero. The moment of a compact translate of a unit highest vector is the corresponding compact translate of its weight. This is the required solution. For scaled labels divide the three Hermitian elements by the scale. $\square$

Lemma 2.6 (Schedules give moments). *A real schedule for (λ, η, ξ) gives a real moment solution for (λ, μ, ξ) , where $\mu = \eta(1)$. No lattice condition is required. The statement also holds for root subsystems with arbitrary orthogonal central coordinates.*

Proof. Apply the three schedule simplifications. The remaining switch gaps, opposite moving-weight pairings, and distances from the time endpoints are strict. Perturb the initial position, shape, and times to rational generic data, allowing the shape endpoint and the final position to vary. Make the shape slopes regular, separate the switch times, and make all ordinary wall crossings simple, transversal, and disjoint from the switches and shape breaks. All strict signs persist.

Project the path to the dominant chamber. At an ordinary crossing, the incoming folded velocity has negative pairing with the crossed simple root, so the direction makes a downward simple Bruhat step. At an original switch, its positive gap makes its root positive in the local chamber frame, and the opposite pairing with the regular moving weight makes its inverse root negative. It too is a downward Bruhat reflection. Shape breaks alone do not change the Weyl frame.

Subdivide at every crossing, switch, and shape break. A common integer dilation makes the semisimple positions and shape increments integral. Each segment then has an extremal intertwiner from Lemma 2.3; its waypoint is the fixed flag of the segment's direction. Their decreasing order permits Lemma 2.4. The resulting nonzero intertwiner gives a real moment solution by Lemma 2.5. Divide by the dilation and pass to the unperturbed labels by compact-orbit closedness. Central coordinates are omitted from the representation argument and restored by their unchanged addition equation; no integrality of a central coordinate is needed. $\square$

2.3 From moments to decreasing real paths

We next establish the real paths on which the area argument operates. During this construction we may freely perturb all three labels before returning to their prescribed values.

For a full flag F and a fundamental weight ω , let $m_\omega(F, H)$ be the least eigenvalue of a Hermitian generator H occurring in a flag vector of weight ω at F . Extend this additively to dominant real weights. A triple of weighted flags (F_j, θ_j) is *strict* if

$$\sum_{j=1}^3 m_{\theta_j}(F_j, H) < 0 \quad (H \in i\mathfrak{k}, |H| = 1). \quad (2.5)$$

Replacing $<$ by $\leq$ defines semistability. We work in the semisimple space in this definition.

Lemma 2.7 (Strict approximation). *Every real moment triple can be approximated by triples of regular labels admitting strict flags. Such strict label triples form an open set. In particular, after choosing a nearby regular shape, there is open freedom to vary its two endpoint labels independently.*

Proof. A moment solution supplies semistable flags: each least supported level is at most its expectation, and the three expectations sum to zero.

For fixed labels, the strict and semistable loci of triples of flags are Zariski open. To see the finite incidence argument, write a test as $H = \text{Ad}(k)d$ with d dominant. A lower bound on a least level means that each relevant highest line lies in the k -translate of an upper level space

$$\bigoplus_{\chi \cdot d \geq c} V_\chi.$$

Such spaces are B -stable, and only finitely many choices occur in the finitely many fundamental representations. For each choice, incidence with a common translating flag is closed; its projection is closed because the flag variety is projective. Among these finitely many incidence types, retain

those for which some dominant d violates the prescribed strict or weak inequality. Their union is exactly the failure locus.

Strict regular data exist. Take regular Hermitian elements X, Y with zero common semisimple Lie centralizer and with $-X - Y$ regular. For example, one can start with X regular toral and choose Y generically with nonzero components on every simple root. Attach the three compact flags realizing these representatives. Equality in the sum of least-level inequalities forces each flag vector to be an eigenvector of the testing generator. For regular weights this makes the generator commute with all three representatives, so it must be zero. The resulting triple is strict.

Intersect the nonempty open flag locus for the original semistable labels with that for this auxiliary strict triple. The product of flag varieties is irreducible, so the intersection is nonempty. At a common triple of flags, a positive convex combination of the original and auxiliary labels is strict and regular. Such labels approach the original labels. Finally, the least supported level is upper semicontinuous in H : disappearing components can only raise its value at the limit. Compactness of the unit sphere makes the strict margin uniform. Small changes of the finitely many weight coefficients preserve that margin. This proves openness and the stated freedom of endpoints. $\square$

The relation between weighted configurations and symmetric-space polygons was developed by Kapovich, Leeb, and Millson [18, Section 5.3, Proposition 5.6]. We give the strict three-point construction needed here, including the inward estimate for the composition of pushes.

Lemma 2.8 (Symmetric-space triangles). *If (λ, μ, ξ^*) admits strict flags, then for every $R > 0$ there is a triangle in $\mathcal{G}/K$ with successive dominant side vectors $R\lambda, R\mu, R\xi^*$.*

Proof. The push of length $R\theta$ from hK toward a full flag F is the path

$$hk \exp(s\theta)K, \quad 0 \leq s \leq R, \quad h^{-1}F = kB, \quad k \in K.$$

It is continuous and well-defined, and the same flag specifies its forward direction all along the push. Compose the three pushes. We prove that this composition has a fixed point.

In polar coordinates $\exp(P)K$, radial distance is $|P|$, and inverse polar coordinates are distance non-increasing. One direct verification uses the differential of the exponential: after translation back by $\exp(-P)$, its Hermitian part is

$$\frac{\sinh(\operatorname{ad} P)}{\operatorname{ad} P} \delta P.$$

This expands lengths and is the identity in the radial direction. Polar decomposition gives global coordinates. In particular, points at a bounded mutual distance and tending to infinity have the same limiting radial unit direction.

At $\exp(lH)K$, $|H| = 1$, the radial derivative of a push toward F is the pairing of H with the compact moment of θ at $\exp(-lH)F$. If $l \rightarrow \infty$ and $H \rightarrow H_0$, then

$$\limsup \frac{d}{ds} |P(s)| \leq m_\theta(F, H_0). \quad (2.6)$$

For a fundamental representation, take a narrow spectral window around the least supported level for H_0 . Before applying the exponential, the flag vector's projection onto a slightly enlarged window has norm bounded below for all nearby H . Multiplication by $\exp(-lH)$ and normalization suppresses levels above that window exponentially. The resulting expectation has limsup at most

the window's upper endpoint. Shrinking the window proves (2.6); addition proves it for real dominant weights.

For fixed R , all three pushes have bounded displacement. Along any sequence tending to infinity with limiting unit direction H_0 , the integrated radial derivatives therefore have total limsup at most

$$R \sum_{j=1}^3 m_{\theta_j}(F_j, H_0) < 0.$$

The derivatives are bounded, so this estimate passes through the finite integrals. It follows, by compactness of unit directions, that every sufficiently large sphere maps strictly inward under the composition. Compose with radial retraction onto the corresponding closed ball and apply Brouwer's fixed-point theorem [6]. A fixed point cannot lie on its boundary, by the strict inward property; an interior fixed point of the retracted map is a fixed point of the original composition. The three pushes from that point form the triangle. $\square$

Lemma 2.9 (Decreasing radial paths). *For generic strict labels and a generic regular polygonal shape ending at μ , there is a dominant path from λ to ξ whose velocities are Weyl translates of the shape velocities, in weakly Bruhat-decreasing order.*

Proof. Move the first vertex of the triangle in Lemma 2.8 to K . Replace its middle side by

$$g_R \exp(R\eta(t))K.$$

Here g_R includes the entering direction frame, so the replacement has the same endpoints. Its initial and final radial Cartan parts are $R\lambda$ and $R\xi$, the latter because the last side is reversed. Denote its radial Cartan part divided by R by $x_R(t)$.

Let $R \rightarrow \infty$. Use all fundamental representations and a regular integral highest-weight representation $V(\sigma)$, with compact-invariant norms. We can choose a common flag z_R such that, in each of these representations and for every nonzero Cartan-weight projection,

$$\|\mathrm{pr}_\chi g_R^\dagger z_R\| \geq c \|\mathrm{pr}_\chi g_R^\dagger\|, \quad c > 0 \tag{2.7}$$

with c independent of R ; use unit flag vectors for z_R . Indeed normalize the finitely many nonzero projected operators to norm one. There are only finitely many possible nonzero-index patterns; for each pattern their tuples range in a compact space. A generic flag vector avoids the kernels of any such tuple, because each nonzero operator is nonzero on some flag vector and a finite union of proper closed subsets does not fill the flag variety. The maximum, over flags, of the minimum of the finitely many norms is positive and continuous in the tuple. Its compact minimum gives c .

The adjoint operation is compatible across these representations. The highest-weight operator norm formula from compact-Cartan decomposition, applied to the adjoint product, and (2.7) show that, after a subsequence,

$$\theta \cdot x(t) = \max_\chi (c_{\theta, \chi} + \chi \cdot \eta(t)). \tag{2.8}$$

Here $c_{\theta, \chi}$ is the limit of $R^{-1} \log \|\mathrm{pr}_\chi g_R^\dagger z_R\|$; values $-\infty$ are allowed. To obtain the formula, the squared norm of $\exp(R\eta(t))g_R^\dagger z_R$ is the sum of the squared norms of its weight projections. The logarithm of its norm therefore differs from the largest logarithm of a projection norm by a bounded amount. The logarithmic error in (2.7) is also bounded before division by R . The coefficients are bounded above by the operator norm of $g_R^\dagger$, whose Cartan part is $R\lambda$. Thus subsequences exist simultaneously for all finitely many coefficients. In each fundamental representation, operator-norm

submultiplicativity applied to $\exp(R(\eta(t) - \eta(s)))$ and its inverse bounds $|\theta \cdot (x_R(t) - x_R(s))|$ by $\max_\chi |\chi \cdot (\eta(t) - \eta(s))|$. The fundamental weights span the Cartan space, so the paths x_R are uniformly Lipschitz. Their fixed endpoints make the limit a dominant path from λ to ξ .

Choose the slopes of η to separate all distinct weight slopes of $V(\sigma)$. Except at finitely many times, the maximum in its version of (2.8) has a unique weight. Every limit of the flags $\exp(R\eta(t))g_R^\dagger z_R$ then lies in that weight space in the regular highest-weight embedding. Such a flag is torus-fixed. The fixed flags have distinct extremal weights, so it is a single fixed flag p on each open interval. The fundamental formulas give $x' = p^{-1}\eta'$ there.

If a later interval gives q , then $p \geq q$. To see this, use a local factorization near p as small U^-U^+ translates of p . Multiplication by the exponential of the scaled regular dominant increment contracts the conjugated U^- factor to the identity. On the compact flag variety this factor acts uniformly close to the identity, even if the remaining U^+ factor diverges. Consequently q belongs to the closure of U^+p , which is exactly the asserted Bruhat comparison. Inversion preserves Bruhat order, so the velocity frames p^{-1} are weakly decreasing as well. $\square$

2.4 Area maximization and the basis of switches

We now refine the decreasing path so that its nonzero changes are single root reflections and their roots form a basis. Area maximization first controls linear dependencies among changes of velocity. The flag-orbit polytope argument then identifies those changes as reflections and proves the cover property needed in the comparison count.

Fix a regular dominant vector ρ and write

$$\mathcal{A}(x) = \int_0^1 \rho \cdot x(t) \, dt.$$

For a fixed regular shape and fixed endpoints, consider the dominant paths supplied by Lemma 2.9, allowing all weakly Bruhat-decreasing direction sequences. This is a compact set: there are finitely many strict direction chains, their ordered time simplices are compact, and the endpoint and chamber conditions are closed. Hence $\mathcal{A}$ has a maximum.

We shall use the following parabolic consequence of the subword criterion to compare directions when a path meets chamber walls. If W_I is a standard parabolic subgroup, replacing an element by the maximal representative of its left W_I -coset preserves Bruhat comparisons. Moreover,

$$u \geq v, \quad u, v \text{ left } I\text{-maximal} \implies gu \geq gv \quad (g \in W_I). \quad (2.9)$$

For completeness, projection to the minimal coset representative preserves comparisons: in a reduced factorization with parabolic prefix, the subword criterion gives the comparison of the remaining minimal representatives. Adding the common length-additive prefix w_I gives the first assertion. Writing $u = w_I \bar{u}$, $v = w_I \bar{v}$ and adding the common length-additive prefix gw_I gives (2.9). Here w_I denotes the longest element of W_I .

At a direction jump $u \rightarrow v$ of such a path, let I consist of the simple walls containing its position. Regularity of the adjacent slopes makes u left I -maximal and v left I -minimal. Reflect the suffix by w_I . The jump is now

$$u \longrightarrow w_I v, \quad u \geq w_I v, \quad (2.10)$$

and both states are left I -maximal. Perform these operations in order, carrying each subsequent instruction by the accumulated left frame. We call this *unfolding*. The original path is recovered

by projection to the dominant chamber. An unfolded jump is *nonzero* if its two states differ; otherwise it disappears. The unfolded terminal point is a fixed Weyl translate of ξ . Endpoint displacements will be computed in this common unfolded space, whereas the area is always that of the folded dominant path.

Lemma 2.10 (Area variation). *For an area-maximizing path with regular shape, the velocity differences at its nonzero unfolded jumps away from shape breaks are linearly independent. Here all differences are measured in the common unfolded coordinate space.*

Proof. An endpoint-preserving variation. Suppose those differences have a nontrivial linear dependence. Move the corresponding jump times linearly in a parameter s in the dependency direction, leaving the shape-break times fixed. For small positive and negative s , the moved times stay in their original slope pieces. The unfolded endpoint change is exactly the sum of the velocity differences times the time changes, so it is zero.

Admissibility after folding. We check that folding these variations still gives admissible Bruhat-decreasing paths. Near an old jump position, only the wall subgroup W_I can be needed for sorting. Both states in (2.10) are left I -maximal. Consequently (2.9) preserves their comparison after any common local sorter $g \in W_I$. Ordinary crossings introduced on either side are downward simple steps. If a moved jump and a wall crossing are simultaneous, sort on the incoming side, make the comparison, then perform the pure crossing on the outgoing direction. To verify the last step, the positive roots of the approached walls pull back under the incoming sorter to positive roots of the local I -system: the incoming slope has negative pairing with them. The other I -maximal state is also negative-facing on that approached subsystem. The outgoing pure sorting is therefore downward. The same reasoning works when a fixed shape break shares the old site. Away from the old sites and breaks there are no new wall contacts; a regular slope cannot run along a wall of a dominant path.

First and second variations of the folded area. We first show that $\mathcal{A}(x_s)$ has a common two-sided derivative at zero. The issue is the change of chamber sorter near old wall sites. There are finitely many jump, break, and wall-contact times. On each adjacent open segment, regularity of the slope makes every vanishing wall coordinate grow at least linearly with distance from its contact time. The unfolded perturbation is uniformly $O(|s|)$. Thus, outside neighborhoods of radius $C|s|$ of those finitely many times, with C fixed sufficiently large, both signs of the perturbation have the same chamber sorter as the original path. They also have the same ordered prefix of moved jumps there. Consequently their unfolded positions have the form $y_s(t) = y_0(t) + sz(t)$, with the same bounded $z(t)$ for both parameter signs, and chamber projection applies the same linear Weyl frame. Chamber projection is globally Lipschitz (it is a continuous, finite piecewise-orthogonal map), so the omitted set, of length $O(|s|)$, contributes only $O(s^2)$ to the area difference. Omitting it from the bounded first-variation integrand also changes that integral by $O(|s|)$. There is therefore a common two-sided first variation, which must vanish at an area maximum.

On either sufficiently small one-sided parameter interval, all resulting sites, including sorting crossings, have affine times $\tau_j(s)$. Indeed positions are affine in (t, s) in each fixed frame, and crossing times solve linear wall equations. Write v_j^-, v_j^+ for the actual folded velocity vectors immediately before and after the j th site. After fixing the one-sided order of these finitely many sites, integration of the piecewise constant velocities with weight $1 - t$ gives

$$\frac{d^2}{ds^2} \mathcal{A}(x_s) = \sum_j \rho \cdot (v_j^+ - v_j^-) (\tau_j'(s))^2. \quad (2.11)$$

A fixed shape break contributes zero. Every other term is nonnegative, because the corresponding comparison is downward and the local shape slope is regular dominant. A moved nonzero original

jump gives a strictly positive term: its two unfolded states lie in distinct W_I -cosets, so sorting cannot erase their difference. The first variation vanishes and the one-sided second variation is strictly positive, contradicting maximality. $\square$

Choose the endpoints generically in the open set supplied by Lemma 2.7, with the regular shape fixed. The independent differences in Lemma 2.10 then form a basis of the semisimple space. Indeed, for any fixed discrete unfolding data, the endpoint balance has the form

$$y(1) - y(0) = b + \sum_j \tau_j(a_j - b_j), \quad (2.12)$$

where the summation is over the nonzero nonbreak jumps, and the fixed vector b includes every contribution from the shape breaks. There are only finitely many discrete data. A deficient span in (2.12) confines the freely varying ends to a proper affine subspace, which generic ends avoid. There are therefore exactly as many such jumps as the semisimple rank, and their times are determined by the balance.

For the same generic choice, every nonzero jump and every shape break lies off all walls in its local frame. For fixed discrete data, translate both unfolded ends by the same small vector. The balance and the basis times stay fixed, while all their positions translate by that vector. Each asserted wall equality is thus a proper affine condition on the free endpoints. Avoid the finitely many such conditions. Pure sorting crossings are not nonzero unfolded jumps and impose no extra restriction here.

2.5 From a basis to cover reflections

The nonbreak jump differences now form a basis, and every nonzero jump site is off the chamber walls. The basis allows us to insert a new direction at any such site and compensate its endpoint displacement by shifting the existing jump times. We use these variations to expose an edge, and then to show that its reflection is a cover.

Lemma 2.11 (Strict exposure). *For such a generic area maximum, let $u \rightarrow v$ be a nonzero jump in local chamber coordinates.*

- (i) *If the site is not a shape break and the slope is d , there is a linear functional f whose maximum among $\{hd : v \leq h \leq u\}$ is attained exactly at ud and vd .*
- (ii) *If it is a shape break with slopes d_-, d_+ , a common functional f has strict maximum at ud_- among the corresponding d_- -vectors, and strict maximum at vd_+ among the corresponding d_+ -vectors.*

Proof. The derivative of the unfolded endpoint with respect to the nonbreak jump times is an isomorphism: its columns are the basis of velocity differences. We can therefore add a linear functional of the unfolded terminal position to the area objective so that all unconstrained basis-time derivatives vanish. This functional is constant on the endpoint constraint, so the constrained maximization problem is unchanged.

At the chosen site, insert an intermediate state h from the Bruhat interval for a small duration, on the left or on the right. Its endpoint displacement is exactly compensated by linear shifts of the basis times. For sufficiently small duration, all these times remain in their fixed slope pieces, so compensation is exact, not merely to first order. If the chosen nonbreak site itself supplies a compensating variable, shift the inserted group as a whole. Chamber feasibility follows from the sorting argument in Lemma 2.10.

Use the fixed Weyl frame at the chosen off-wall site to identify its local coordinates with the common unfolded space. The first variation of the modified objective is then a linear functional f of the inserted local velocity difference. It is the same from either side of the site: it integrates the effect of that displacement on the later folded path, together with the terminal multiplier, and changing the lower time endpoint by a vanishing interval does not change the first variation. At a nonbreak site, shifting the original jump has vanishing first variation, so $f(ud) = f(vd)$. Maximality gives $f(hd) \leq f(ud)$. At a break, a left insertion gives $f(hd_-) \leq f(ud_-)$ and a right insertion gives $f(hd_+) \leq f(vd_+)$. These include extension of the opposite endpoint state across the break.

For an inserted interval parameterized by its duration, the two boundary-time derivatives differ by one, so compensation cannot fix both. Thus each genuine intermediate insertion, or opposite-state extension at a break, moves at least one nonzero jump time. If its first-order inequality were equality, Equation (2.11) would give a strictly positive one-sided second variation and hence an improvement. Therefore every such inequality is strict. This proves both assertions. $\square$

The orbit-closure polytope and its root-parallel edges are classical: see Atiyah [1, Theorem 2] and Gel'fand–Serganova [13, Section 7.1, Theorem 1]. We include the proof in the form needed here, keeping track also of the fixed-flag labels as a regular real highest weight varies.

Lemma 2.12 (Flag orbit polytopes). *Choose a flag $b \in U^+u \cap U^-v$, with $u \geq v$. The torus-fixed points of its torus-orbit closure lie in $[v, u]$ and include u, v . For a regular dominant real weight d , take the convex hull of the corresponding vectors hd . Its normal fan, with vertices labeled by fixed flags, is independent of d . Every edge joins two vertices related by a root reflection.*

Proof. The orbit closure is contained in the Schubert closure through u and the opposite Schubert closure through v , giving the interval assertion; the two flow limits supply its endpoints. For integral regular d , this is the weight polytope of the torus orbit of the line of the flag vector in $V(d)$. Vertices are obtained by generic one-parameter limits, and the fixed flags in this embedding have Weyl-extremal weights.

The flag vector for a sum of fundamental weights is the tensor of their flag vectors in the Cartan summand. Its torus weight polytope is therefore the corresponding Minkowski sum of the fundamental polytopes. For strictly positive coefficients its normal fan is their common refinement, independent of those coefficients. The labels are independent as well, since a generic torus limit is the same flag in all highest-line maps. Positive real coefficients give the same conclusion for real regular d .

For integral d , an edge is exposed by a one-parameter limit and comes from a one-dimensional torus orbit in the flag variety. Its codimension-one stabilizing subtorus annihilates a tangent root at a fixed endpoint, since the tangent weights of the flag variety there are roots. Thus the edge direction is a root direction. Its two Weyl vertices have equal norm, and two distinct equal-norm points on a line in root direction are exchanged by the corresponding root reflection. The common labeled fan carries this statement to every regular real d . $\square$

Lemma 2.13 (Exposed jumps are covers). *The generic area maximum has no nonzero jump at a shape break. Every other nonzero unfolded jump is a reflection and is a Bruhat cover in its local chamber frame.*

Proof. Choose the flag in Lemma 2.12. At a shape break, Lemma 2.11 would give two different unique maximizing vertex labels for the same functional on two polytopes with the same labeled

normal fan. This is impossible. At a nonbreak site, that lemma exposes exactly the endpoint vertices, so they form an edge. Lemma 2.12 makes the jump a reflection.

Write it as $u \rightarrow s_\beta u$, with $\beta > 0$ and $u^{-1}\beta < 0$. Suppose it is not a Bruhat cover. In these simply laced systems there are positive roots α, δ such that

$$\beta = \alpha + \delta, \quad u^{-1}\alpha < 0, \quad u^{-1}\delta < 0. \quad (2.13)$$

Here is the inversion count proving the assertion. Apart from β , the positive roots taken negative by s_β are paired as $\alpha, \beta - \alpha$. A pair with both roots inverted by u^{-1} contributes two to the length drop, whereas a mixed pair contributes zero. A pair with neither root inverted is impossible, since their sum has negative inverse image $u^{-1}\beta$. Thus a drop greater than one forces (2.13).

In the rank-two subsystem on these three positive roots, both $s_\alpha u$ and $s_\delta u$ are proper intermediate states between u and $s_\beta u$. Strict exposure therefore forces $f \cdot \alpha < 0$ and $f \cdot \delta < 0$: for example, $s_\alpha u d - u d = -(u d \cdot \alpha)\alpha$ is a positive multiple of α . Equality of the two exposed endpoint values gives $f \cdot \beta = 0$, contrary to (2.13). The reflection is a cover. $\square$

Proof of Proposition 2.2. First use generic strict labels and a generic regular shape. Lemma 2.9 supplies the nonempty compact path class. At an area maximum, Lemma 2.10 and the generic endpoint balance give a basis of nonbreak velocity differences; Lemma 2.13 removes all nonzero break jumps and makes every remaining nonzero jump a cover reflection. Its velocity difference is a nonzero scalar multiple of its root. Thus those roots form a basis. In the unfolded coordinates they obey (2.1): the local comparison root has nonnegative position pairing and negative pairing with the incoming moving weight. They therefore give a schedule.

For the prescribed moment data, use Lemma 2.7 and regular shapes converging to the prescribed shape. Such shapes can be obtained by small positive interpolation with shapes ending at the perturbing regular weights, followed by generic choices of their slopes and break values. At each stage the strict set gives the open endpoint freedom required in the area argument. The switch roots, frames, and endpoint Weyl elements take only finitely many values; pass to a subsequence on which these discrete data are fixed. The ordered switch times lie in a compact simplex. Uniform convergence of the shapes then gives convergence of the positions and switch coefficients. The roots remain the same independent basis, and the signs become weak. Central coordinates have been carried unchanged by addition throughout. This proves the stated limiting schedule. $\square$

We record the concrete consequence of the cover property needed in the integrality argument. For type D , use signed labels a with $y_a = \pm y_{|a|}$ and likewise for the moving values $w\mu$.

Corollary 2.14 (No strict third-axis obstruction). *At a switch of the limiting schedule in Proposition 2.2, suppose the two exchanged signed labels have strict opposite position and moving-value orders. No signed label on a third physical axis can lie strictly between them in both orders. The moving values are read immediately before the switch, including its specified position in a simultaneous cluster.*

Proof. Such strict inequalities would persist in the regular approximating schedules. In their local sorted frame the third label would give the two positive roots in (2.13), both inverted by the incoming direction. The switch would then have Bruhat length drop greater than one, contrary to its cover property. $\square$

3 Lattices and continuation of schedules

A real basis of roots need not be an integral basis of the root lattice. Thus the endpoint balance of the basic schedule does not by itself make its switch coefficients integral. We identify the type- D obstruction and construct a deformation that removes it when a suitable local modification is available. Section 4 will force such a modification.

The result concerns the switch condition in Definition 2.1, not the stronger chain condition in a highest-weight path model. An integral chain in the Lakshmibai–Seshadri model for a dominant weight θ must be saturated in the Bruhat order on W/W_θ [26, Section 2.2]; the distinction also appears in [19, Theorem 5.6]. The passage from integral switches to representations remains separate.

Theorem 3.1 (Integral schedules). *Let the three labels have integral root pairings and satisfy $\lambda + \mu - \xi \in Q$, and let η be any polygonal shape with dominant increments and endpoint μ . A real moment solution gives an integral schedule in either of the following cases:*

- (i) *the root system is a permutation system;*
- (ii) *the root system is D_l , and μ has at most one zero coordinate.*

An orthogonal central space is allowed in either case; its equation is ordinary vector addition.

For a permutation system, a root basis is a spanning tree in each semisimple factor and is an integral basis of its root lattice. Proposition 2.2 and the endpoint balance therefore prove the first assertion. We prove the second by induction on l . Throughout this section, uses of Theorem 3.1 in orthogonal blocks refer only to strictly smaller ranks.

3.1 Blocks, parity, and the exceptional graph

Associate a signed graph on the coordinate axes to a collection of D_l -roots. An edge $e_i - \varepsilon e_j$, with $\varepsilon \in \{1, -1\}$, imposes the signed identification $x_i = \varepsilon x_j$. A connected component is *balanced* if these identifications are consistent around every cycle. Changing the signs of its axes then makes its roots a permutation system. An unbalanced component has full coordinate span. This is the signed-graphic representation of the classical root systems studied by Zaslavsky [36, 37]. We will also need its integer-lattice information, which is proved below by leaf removal and the cycle equations.

Lemma 3.2 (Combining blocks). *Suppose a moment solution splits into commuting coordinate blocks of permutation type and orthogonal type. Assign the initial, moving, and final coordinates using this splitting. Under the hypotheses of Theorem 3.1(ii), integral schedules can be combined from the blocks. In particular, induction applies if there is a permutation block, if there are at least three proper orthogonal blocks, or if there are two proper orthogonal blocks of even parity. Here the parity of an orthogonal block is the sum, modulo two, of its assigned difference $\lambda + \mu - \xi$.*

Proof. We first explain the assignments and their arithmetic. In a schedule whose roots lie in the stated blocks, the initial moving frame assigns shape coordinates to each block; the endpoint assigns the final coordinates. Restricting the schedule gives a moment solution in the block by Lemma 2.6. Conversely, after sorting the assigned moving weight for the block, all restricted shape increments are dominant in the same frame. Indeed every root comparison has the same weak sign on every dominant increment as on their sum. The orthogonal central coordinates are simply

retained. Schedules obtained in the blocks can thus be superposed. Their root inequalities are internal to the blocks, so no comparison with an outside coordinate is required.

The same assignments are available for a matrix splitting. Choose an oriented adapted orthonormal frame, and then choose representatives of each of the three matrices in its blocks. The assembled representatives belong to the original special-orthogonal orbits. For the negative end matrix, negate its representative to obtain the assigned end label for addition. Subgroup Weyl actions on these assignments are permitted in the full Weyl coset; a different initial representative can finally be returned to λ by a common Weyl action on the path.

Each type- D weight has a uniform coordinate fractional part, zero or one-half, invariant under signed permutations. The global root-lattice condition consequently makes every assigned coordinate difference an integer. On a permutation block, the central equation gives zero signed sum. Its difference lies in the permutation root lattice and has even ordinary sum as well. On an orthogonal block, changing allowed representatives changes the difference by a block root, so its parity is well-defined. The number of odd orthogonal blocks is even.

Even orthogonal blocks can be solved separately by induction; odd ones can be joined in pairs, using their direct-sum moment solution in the larger orthogonal block. If there are at least three orthogonal blocks, every such union is still proper. If there is a permutation block, all orthogonal blocks together occupy a proper subspace and have even total parity; solve that union and the permutation blocks separately. A single permutation block on all axes is covered by the first assertion of the theorem. A real two-dimensional orthogonal block has only its central equation and is even. Finally, a restriction of the moving coordinates retains the at-most-one-zero hypothesis. $\square$

One useful source of a permutation block is a nonzero common commuting real skew-symmetric operator. Its kernel is an orthogonal block. On each subspace belonging to a nonzero absolute eigenvalue, its normalized restriction is a complex structure; the commuting skew matrices become Hermitian matrices and give a permutation-type addition problem. Thus Lemma 3.2 applies to this splitting, also inside an already assigned orthogonal block.

Lemma 3.3 (The exceptional case). *If no integral schedule exists, any real schedule with at most l switches has exactly l independent switch roots, and its graph has exactly two connected components, both unbalanced and of odd parity. Each component is unicyclic. On its cycle the switch coefficients are half-odd-integers, and off the cycle they are integers.*

Proof. If the roots have deficient span, some connected component is balanced, possibly an isolated axis. Lemma 3.2 applies. Otherwise there are exactly l independent roots. Each component is then unbalanced and has as many edges as vertices, hence has a single cycle.

The roots of an unbalanced unicyclic component generate precisely its even-sum integer lattice. To see this, remove leaves successively. Their coefficients are determined integrally by the corresponding coordinate equations. The remaining cycle equations determine all cycle coefficients from one of them, with integral differences or sums; inconsistency of the signs closes the final equation with a factor two. The resulting determinant has absolute value two, so the image is the even-sum lattice. The coefficients are integral when the right-hand side has even coordinate sum and all half-odd-integral on the cycle when its coordinate sum is odd. This argument includes an unbalanced cycle with two edges.

A connected graph therefore already gives integral switch levels. Multiple components can be treated by Lemma 3.2, except for exactly two odd orthogonal components. These are the only possibility under the assumed failure of integral realizability. $\square$

In particular the induction starts in every rank too small to contain two unbalanced components. Notice also that the reduction applies to any real schedule with at most l switches, not just to the original basis schedule. For example, a connected balanced graph with l edges gives a permutation reduction, although it is not a root basis.

Lemma 3.4 (Closedness and generic shapes). *For fixed labels, the existence of an integral schedule is closed under uniform convergence of shapes with dominant increments. It suffices to prove Theorem 3.1 for polygonal shapes whose slopes are relatively interior to the smallest chamber face containing μ , with generic break values in that face.*

Proof. Delete zero-total, endpoint-time, and zero-gap switches by the simplifications in Definition 2.1. These operations preserve integrality. For the remaining strict switches, a small regular perturbation and chamber projection gives a strictly downward Bruhat step at each switch. Their number is therefore at most the length of the longest Weyl element. This bound is independent of the shape. There are finitely many ordered root lists and initial frames; after taking a subsequence they are constant and their times converge. The switch levels are bounded integers, hence are also constant on a further subsequence. Passing to the limit preserves the endpoint, weak signs, and these integer levels.

Let F be the smallest chamber face containing μ . Every increment of every allowed shape lies in F : a simple-root pairing which vanishes on μ is a sum of nonnegative increment pairings, each of which must vanish. Slight interpolation with the straight shape $t\mu$ makes all slopes relatively interior to F . Subdivision and perturbation of break values, keeping the endpoint, then impose any finite collection of open dense genericity conditions. The precise conditions used below are given in Section 4. $\square$

We may now assume that there is no integral schedule for such a shape. Take the schedule from Proposition 2.2. Lemma 3.3 gives two odd components, denoted P, Q . Every switch has positive gap and positive total and lies at an interior time: deleting a zero sign or a boundary switch would leave at most $l - 1$ switches. Retain also the limiting cover property of this particular schedule. It will be used only in the counting argument, not as a condition on its later deformations.

3.2 The half-relation deformation

We seek to replace a cluster of the exceptional schedule by a cluster with one additional switch. Outside that cluster the modification must be recoverable by undoing a suffix reflection or a change of moving frame that fixes the shape. First consider the lattice part of this problem: an exact local modification with $k = l + 1$ ordered roots generating $Q(D_l)$. Such a list need not retain all the old roots. We will move its coefficients to integers without changing its endpoint; the local-link conditions below will supply the feasibility needed to begin that motion. Write

$$p_i = s_{\beta_i} \cdots s_{\beta_1}, \quad p_0 = 1, \quad \gamma_i = p_{i-1}^{-1} \beta_i, \quad b = w_{\text{in}} \eta.$$

Thus γ_i is the switch root in the initial velocity frame, and $m_i = -\gamma_i \cdot b(t_i)$. The two endpoint balances are

$$y(1) = \lambda + p_k b(1) - \sum_{i=1}^k m_i \beta_i, \tag{3.1}$$

$$p_k^{-1} y(1) = \lambda + b(1) - \sum_{i=1}^k G_i \gamma_i, \tag{3.2}$$

where

$$G_i = \lambda \cdot \beta_i - \sum_{j < i} m_j \beta_j \cdot \beta_i - m_i. \quad (3.3)$$

The first identity is the schedule balance; the second follows by reflecting the position back at every switch. Formula (3.3) also shows that the change from m to G is affine integral and unimodular, with diagonal coefficients -1 .

Lemma 3.5 (Half-kernel bounds and potential). *Suppose the $l + 1$ roots generate $Q(D_l)$, and their exact base coefficients m^0 are half-integral. If m^0 is not already integral, either orientation d of half the primitive integer relation among the roots satisfies*

$$m^0 + d \in \mathbb{Z}^{l+1}, \quad |d_i| \leq 1, \quad |d_i^G| \leq 1, \quad (3.4)$$

where d^G is its image under the linear part of (3.3). Moreover, there is a vector L such that

$$d_i = L \cdot \gamma_i \quad (1 \leq i \leq l + 1). \quad (3.5)$$

Proof. A full-rank signed graph with c unbalanced components generates an integer lattice of index 2^c . Thus a graph generating $Q(D_l)$ is connected and unbalanced. Deleting one edge leaves at most two components. Each nonzero maximal minor consequently has absolute value two or four: its components are unbalanced unicyclic graphs. Their gcd is two, the index of $Q(D_l)$ in $\mathbb{Z}^l$. Cofactors therefore give a primitive kernel vector with entries of magnitude at most two.

The velocity roots generate the same lattice as the position roots: γ_i equals β_i plus an integral combination of earlier roots, with a triangular inverse of the same form. Applying the preceding minor argument in that frame, and using the unimodular coefficient change (3.3), proves both derivative bounds. The endpoint right-hand side in (3.1) lies in $Q(D_l)$, so it has an integer coefficient vector. Its difference from m^0 is a multiple of the primitive relation. Since that relation has relatively prime entries and m^0 is half-integral but not integral, the multiple is half-odd-integral. Adding either half relation makes all coefficients integral.

Finally, telescoping the reflections gives

$$\sum_i (L \cdot \gamma_i) \beta_i = (1 - p_k) L. \quad (3.6)$$

The determinant of p_k is $(-1)^{l+1}$. An orthogonal operator in dimension l without eigenvalue $+1$ has determinant $(-1)^l$, by pairing its nonreal eigenvalues. Hence p_k has a nonzero fixed vector. Since the γ_i span, its pairings with them give a nonzero relation. The relation space is one-dimensional; scaling this fixed vector proves (3.5). $\square$

We vary the coefficients by

$$m(s) = m^0 + sd, \quad 0 \leq s \leq 1. \quad (3.7)$$

At parameter one every coefficient is integral, and the position balance keeps the endpoint fixed. This line of coefficient vectors is not yet a family of schedules. The time for its i -th switch must solve

$$-\gamma_i \cdot b(t_i(s)) = m_i(s).$$

A positive total makes the left side strictly increasing along our relatively regular shape, so it has a local inverse at an interior base time. These inverse times are candidates: they need not occur in

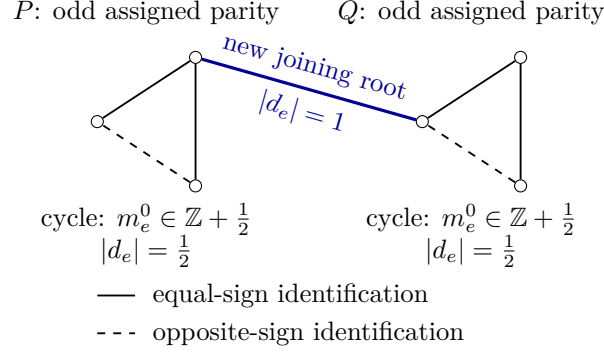


Figure 1: The graph underlying the half-relation deformation. The two original components are unbalanced and have odd assigned difference parity. A joining root connects their even-sum lattices inside the full type- D root lattice. For the displayed graph, half the primitive relation has absolute value $\frac{1}{2}$ on the cycles and 1 on the joining edge. More generally the latter value occurs on the whole connecting path, and coefficients on other attached trees vanish. Cycle lengths and omitted branches are schematic; local feasibility is supplied by a local link.

the prescribed order, and their position gaps must remain nonnegative. We must start the motion at the modified cluster and then continue it through collisions without losing these conditions.

Here is the exact form of the local input we shall use. A *local link* modifies the original schedule at one time, replacing some of its switches by one more switch in total, such that:

- (a) the new $l + 1$ roots generate $Q(D_l)$, and the exact base coefficients are half-integral;
- (b) one orientation of d in (3.7) gives admissible, weakly ordered candidate switches within the modified cluster for all sufficiently small positive s , allowing commutations of disjoint switches;
- (c) at the other time clusters the modification can be forgotten, either by a common reflection of the suffix or by a discrepancy of moving frames that fixes the entire shape.

In the latter alternative, equality of assigned moving weights is sufficient. Indeed W_μ fixes the face F in Lemma 3.4 pointwise. If $w'\mu = w\mu$, then $w^{-1}w' \in W_\mu$, and therefore

$$w'\eta(t) = w\eta(t) \quad \text{for every } t. \quad (3.8)$$

Thus a local reassignment of equal moving values preserves every later shape value, not just the value at the modified time.

Figure 1 shows the lattice calculation in the simplest joining-root case. Its half-relation reaches integer coefficients; the local-link conditions are what permit those coefficients to be realized by nearby switch times.

3.3 Sorting collisions and continuing a link

Proposition 3.6 (Continuation). *Under the standing assumption of no integral schedule, a local link cannot exist.*

Proof. Continuation may replace the ordered root list when times collide. We therefore follow coefficient lines in a finite set of root lists, not one fixed list. Each replacement preserves the

endpoint and transforms the extrapolated coefficients at parameter one by an integral unimodular change. First we exclude removable switches at an interior parameter. We then use a common potential to select compatible times for alternative routes through a tied cluster. The local-link conditions start the motion at parameter zero, and compactness carries it to parameter one.

Strictness at a continuation point. Suppose an exact augmented schedule occurs at a parameter $0 < s_* < 1$, with a primitive half-kernel direction and integral extrapolation at one. If a switch is at time zero or one, its coefficient is integral; a zero-total switch has coefficient zero. Consequently

$$(1 - s_*)d_i = m_i(1) - m_i(s_*) \in \mathbb{Z}, \quad |(1 - s_*)d_i| < 1,$$

so $d_i = 0$. Deleting that switch leaves a nonzero relation among at most l roots, hence deficient rank. At an endpoint one may delete the entire endpoint cluster. Either operation contradicts Lemma 3.3.

If a gap is zero, the same reasoning applied to G_i gives $d_i^G = 0$, since the extrapolated gap is integral by (3.3). Reflect the suffix and delete the switch. All remaining velocity roots are unchanged: reflection changes both their position labels and the accumulated frame by the same factor, which cancels in $p_{i-1}^{-1}\beta_i$. The residual dependence again gives deficient rank. Thus all switches at such a parameter are interior, with strict gaps and totals.

At $s_* = 0$ this arithmetic would not exclude a nonzero half-kernel entry. Instead, in an *unaffected* cluster use the forgetting property of the local link. Any reordering there preserving its full reflection product and exact displacement can also be performed after forgetting the augmentation. In case (3.8), the same fixed right stabilizer discrepancy persists after the reordering; in the reflected-suffix case conjugate the reordered roots back. One obtains an exact schedule with the original l switches. A zero sign or boundary switch could then be removed, leaving at most $l - 1$ switches, again impossible. This argument applies after every successive reordering at that base time.

A common potential at a collision. Suppose several candidate times coincide at τ when $s = s_*$. By Lemma 3.5, throughout that cluster

$$m_i(s) = \gamma_i \cdot (-b(\tau) + (s - s_*)L). \quad (3.9)$$

The candidate time belonging to a root line γ therefore solves

$$\gamma \cdot (b(t) - b(\tau) + (s - s_*)L) = 0. \quad (3.10)$$

It depends only on that root line and the common potential, not on the route chosen through the cluster.

Commute an adjacent pair with orthogonal roots whenever its candidate times are inverted. For a nonorthogonal pair, the rank-two subsystem is A_2 . At the base point choose one regular refinement of the common position order and one of the incoming velocity order, preserving their strict inequalities, and transport the latter through the two reflections. The two steps are decreasing in this single rank-two Bruhat order. Both are covers: their length drops are positive odd integers whose sum is at most three, hence their sum is two. Every length-two interval in the A_2 Bruhat order is a diamond. Its other route therefore also consists of two downward covers: the position root is positive, and its inverse image under the current velocity state is negative, at each step. Forgetting the common regular refinements gives weakly correct position and velocity signs for this other route. Identical consecutive reflection lines cannot occur, since the first would reverse the strict velocity inequality required by the second. Replace the pair by that route with

the same product. The preceding deletion arguments exclude equality before we require its nearby inverse times.

The displacement of the pair is unchanged by (3.9) and the telescoping identity (3.6). Any two nonparallel roots in this A_2 subsystem form a basis of its root lattice. Thus the old and new coefficient pairs are related by an integral unimodular map. Outside the pair the accumulated frame, roots, and coefficients are unchanged. It follows that the full lattice, the primitive half-kernel direction, and integral extrapolation at one are all preserved.

Termination of sorting. Let i, j denote the simple root lines of the velocity-positive system and $h = i + j$ its remaining root line. Their candidate times are T_i, T_j, T_h . Choose representatives consistently with the incoming moving order, so that their level functions have the same strict monotonicity. Reversing a representative does not change its candidate-time equation. The left-hand side of (3.10) for h is the sum of those for i, j ; strict monotonicity implies that T_h lies strictly between T_i, T_j when they differ. This holds also across a shape break. The four possible diamonds, written as successive right reflection labels in the velocity frame, are

lengths	alternative routes
$3 \longrightarrow 1$	$ij \longleftrightarrow jh$
$3 \longrightarrow 1$	$ih \longleftrightarrow ji$
$2 \longrightarrow 0$	$ji \longleftrightarrow hj$
$2 \longrightarrow 0$	$ij \longleftrightarrow hi$

Indeed, writing $s_h = s_i s_j s_i = s_j s_i s_j$, their products are $s_i s_j = s_j s_h = s_h s_i$ and $s_j s_i = s_i s_h = s_h s_j$; the starting length determines which two factorizations form the interval. These are the same common position and velocity refinements used above. In each row, if the current route is descending in candidate time, the other route has strictly earlier first time. For example, in the first row $T_i > T_j$ makes the route jh start earlier; conversely $T_j > T_h$ implies $T_j > T_h > T_i$, so ij starts earlier. The other three rows follow by the same between-time inequality. Orthogonal commutation also strictly decreases the first time of the inverted pair.

The full candidate-time list therefore decreases lexicographically. Only finitely many root lists are possible, and each root line has its fixed candidate time from (3.10). Sorting terminates in a weakly ordered list. All its strict gaps and totals persist in a common sufficiently small neighborhood of the base parameter, since there are only finitely many possible lists. At $s_* = 0$ apply this procedure only to unaffected clusters; the modified cluster is feasible by the definition of a local link. Consequently the link gives a genuine schedule for small positive s .

Arrival at the integral endpoint. Consider all parameters in $(0, 1]$ admitting an augmented schedule with $l + 1$ lattice-generating roots, a primitive half-kernel direction, and integral extrapolation at one. This set is nonempty by the preceding paragraph. Root lists, initial frames, and endpoint representatives range over finite sets. The actual coefficients are bounded by the compact shape, and every direction coordinate has absolute value at most one. Their extrapolated integer values at one are therefore bounded as well and range over a finite set. The half-kernel directions themselves also range over a finite set.

At the supremum, choose a subsequence fixing all this discrete data. The ordered switch times have a convergent subsequence; endpoint balance and weak signs are closed conditions. Thus the supremum is feasible. If it is less than one, the strictness argument excludes all boundary or removable switches there, and the common-potential sorting extends feasibility to its right. This

is impossible. Parameter one is attained, and every coefficient is then integral, contrary to the standing assumption. $\square$

The proof only asks for weak order inside a feasible modified cluster. If two new transfers have the same strictly increasing level function and equal coefficient derivatives, their times remain exactly equal; the specified order of simultaneous switches is permitted.

Lemma 3.7 (Direct links). *In the exceptional schedule with components P, Q , each of the following produces a forbidden local link:*

- (i) *a signed cross-component pair with strict concordant position and moving-value orders at time zero;*
- (ii) *a signed cross-component pair with strict discordant orders at time one;*
- (iii) *a position crossing of a cross-component pair with distinct moving values between old switch times;*
- (iv) *the crossing in (iii) at an old switch time, if both axes are unused by its cluster.*

Proof. In the first case insert the cross swap at time zero, choosing the initial frame so that the old frame occurs just after the swap. Before the swap the velocity order is the strict inverse of its position order, so the gap is positive. Its coefficient is zero. The position graph has gained a bridge joining the two odd unbalanced components; its roots generate the full root lattice. In the primitive relation the bridge coefficient has magnitude two. Indeed, prune the off-cycle branches and follow the remaining connecting path: the cycle equations give equal coefficient magnitudes around each odd cycle, and their sign inconsistency makes the connecting flux twice that magnitude. Thus the primitive magnitudes are one on the cycles and two along the connecting path. The half-kernel derivative on the bridge has magnitude one. Choose its sign to make the new time move into $(0, 1)$. All old gaps remain positive. The inserted edge can plainly be forgotten at the exact base time. The end case is the same construction with the old state before the new swap; choose the derivative to move its time leftward from one. Its base coefficient is the integral total root pairing.

For a crossing, insert the swap with its strict approaching inversion and reflect the entire old suffix. At the base point its gap is zero, its total is positive, and continuity holds because the crossing position is fixed by the reflection. In the velocity frame all old roots are retained. The two component root bases in that frame still generate their respective even-sum lattices, and the new root is a bridge between them. Hence the augmented roots generate $Q(D_l)$, and the primitive half-kernel direction in the gap coordinates has nonzero, indeed unit, coordinate on the bridge. Choose its sign so that the new gap becomes positive. Its time remains interior. Formula (3.3), with zero base gap, shows that the inserted coefficient is half-integral; the other base coefficients remain half-integral. Forgetting is precisely reflection of the suffix back.

If the crossing lies at an old cluster unused by its two axes, its new root is orthogonal to all roots in that cluster. Commute these switches to put their nearby candidate times in order. This proves local feasibility also in the last case. Proposition 3.6 excludes each construction. $\square$

It remains to show that the exceptional schedule necessarily admits a link. Proposition 4.2 supplies a strict excess of cross-component comparisons, while Proposition 4.4 shows that this excess cannot disappear without a locally feasible link. These two assertions, proved next, complete Theorem 3.1.

4 A strict comparison count and simultaneous switches

We complete the proof of Theorem 3.1. Work under its rank induction and suppose, for a contradiction, that no integral schedule exists. By Lemma 3.3, the basic schedule has l independent switch roots and exactly two components, denoted by P, Q , both unbalanced and of odd parity. Every switch has positive gap, positive total, and an interior time. The limiting cover property of Proposition 2.2 is retained. Any realization with fewer than l switches, or with at most l switches outside this exceptional configuration, contradicts Lemma 3.3. Every feasible extra-edge link contradicts Proposition 3.6.

Write $\hat{P} = \{\pm i : i \in P\}$ and similarly for Q . For a signed label a , put $e_{-a} = -e_a$, $y_a = y \cdot e_a$, and $v_a = (w\mu) \cdot e_a$ in the current frame. A cross-comparison means a pair $(a, c) \in \hat{P} \times \hat{Q}$. We count each with weight $1/2$, so that a comparison and its signed mirror together have weight one. Set

$$r = 2|P|, \quad h = 2|Q|, \quad R = \frac{rh}{2}.$$

Let p_λ, p_μ, p_ξ be the numbers, with this same weight, of cross ties in the initial positions, assigned moving values, and final positions. The middle number is independent of the time at which the assigned values are read.

4.1 The strict moment inequality

Our aim is the strict inequality $R > p_\lambda + p_\mu + p_\xi$. The moments on the two components turn its tie counts into spectral multiplicity counts. For a Hermitian triple on $\mathbb{C}^r$, its *profile* is the collection $m = (m_i(a))_{i,a}$, where $m_i(a)$ is the multiplicity of the numerical eigenlevel a of its i -th member. Absent levels have multiplicity zero. Profiles of different sizes are compared using the same three sets of numerical levels.

Writing m^P, m^Q for the profiles of the component moments, we will have $p_\lambda + p_\mu + p_\xi = (m^P \cdot m^Q)/2$. An orbit-dimension estimate will give the required weak bound by Cauchy–Schwarz, with a separate treatment of a moving zero level. Without that zero, equality forces proportional profiles of full norms r, h . We will also establish irreducibility in this equality case, so the next lemma forces the two sizes to agree. In the proof of strictness, identical profiles will yield a forbidden permutation block.

The lemma’s direct-sum argument is a flag formulation of the canonical-decomposition methods for quiver representations developed by Kac and Schofield [16, 34]. We give it directly in terms of the three spectral filtrations.

Lemma 4.1 (Profile rigidity). *Suppose two irreducible Hermitian zero-sum triples have proportional profiles m, m' , of sizes r, h , respectively, and*

$$\|m\|^2 = r^2, \quad \|m'\|^2 = h^2.$$

Then $r = h$ and $m = m'$.

Proof. Flags with scalar endomorphisms. For a profile m , let $\mathcal{F}(m)$ be the product of the three partial-flag varieties with the prescribed upper-level dimensions. Its complex dimension is

$$\dim_{\mathbb{C}} \mathcal{F}(m) = \frac{3r^2 - \|m\|^2}{2} = r^2. \quad (4.1)$$

The product of the corresponding Hermitian orbits has real dimension $2r^2$. At an irreducible zero-sum triple, the sum map is submersive onto the traceless Hermitian matrices: the orthogonal

cokernel consists of common commuting Hermitian matrices, which are scalar. Its local zero-sum fiber therefore has dimension $r^2 + 1$, whereas simultaneous unitary conjugation has orbit dimension $r^2 - 1$. Irreducibility is open. It follows that these fixed spectra admit arbitrarily many pairwise unitarily inequivalent irreducible zero-sum triples.

Their upper-level flags have scalar endomorphisms, and flags from two inequivalent triples have no nonzero morphism. Indeed, let L preserve all three upper-level filtrations, from a triple (X_i) to a triple (X'_i) . In spectral coordinates its nonzero matrix entries go from a level a only to levels $a' \geq a$. Consequently

$$\mathrm{tr}(L^* X'_i L - L^* L X_i) = \sum_{a, a'} (a' - a) \|L_{a', a}\|^2 \geq 0.$$

The sum of these three quantities is zero. Thus every nonzero entry has $a' = a$, and $X'_i L = L X_i$ for all i . Irreducibility now gives the assertions. Here an invertible complex intertwiner between irreducible Hermitian triples is a scalar multiple of a unitary intertwiner, since $L^* L$ commutes with the source triple.

Direct sums fill an open set. The Hermitian triples have supplied Hom-free spectral flags. We now work in the complex flag varieties: the configurations in the direct-sum families need not themselves arise from Hermitian zero-sum triples. For every positive integer b , direct sums of b configurations in $\mathcal{F}(m)$ having scalar endomorphisms, with arbitrary changes of frame, contain a nonempty Zariski-open subset of $\mathcal{F}(bm)$. To prove this, choose b pairwise Hom-free configurations as above and examine the differential of the direct-sum map, including the action of $\mathrm{GL}(br, \mathbb{C})$. The flag tangent space splits into ordered blocks indexed by the summands. Its diagonal blocks are supplied by variation of the individual flags. Each off-diagonal ordered block has dimension r^2 : the contribution of its i -th flag is

$$\sum_{a < b} m_i(a) m_i(b) = \frac{r^2 - \sum_a m_i(a)^2}{2},$$

and summing gives (4.1). Change of frame maps a space of $r \times r$ matrices to that block; its kernel is exactly the space of filtration-preserving maps between the two summands, which is zero. Hence this differential is surjective. The direct-sum map is therefore dominant. Chevalley's theorem makes its image constructible [35, Tag 054K]; a dense constructible subset of an irreducible variety contains a nonempty Zariski-open subset. Requiring scalar endomorphisms on the summands is an open condition and includes our chosen point, so the same assertion holds with this requirement.

We finish with the scalar-endomorphism case of the Krull–Schmidt uniqueness argument; compare [2, Theorem 1]. Choose positive integers b, b' with $bm = b'm'$. The two nonempty open subsets just obtained intersect in the irreducible variety $\mathcal{F}(bm)$. One configuration in their intersection admits two direct-sum decompositions

$$V = \bigoplus_i V_i = \bigoplus_j W_j,$$

whose summands have scalar endomorphisms, with dimensions r and h , respectively. Inclusions and projections preserve all three flags. On a fixed V_i , the sum of the composites $V_i \rightarrow W_j \rightarrow V_i$ is the identity. One composite is therefore a nonzero scalar. Write it as $gf = c \mathrm{id}_{V_i}$, $c \neq 0$. The endomorphism fg of W_j is also scalar, and $(fg)f = cf$ shows that its scalar is c . Thus f is an isomorphism, so $r = h$. Proportional profiles of equal size are equal. $\square$

Proposition 4.2 (Strict comparison count). *In the standing exceptional configuration,*

$$C := R - p_\lambda - p_\mu - p_\xi > 0. \quad (4.2)$$

Proof. Choose skew-symmetric zero-sum triples from the separate moment solutions in P and Q , supplied by Lemma 2.6. Let m^P, m^Q be the profiles of i times these triples. Since the spectra of a real skew matrix occur in opposite pairs,

$$p_\lambda + p_\mu + p_\xi = \frac{m^P \cdot m^Q}{2}. \quad (4.3)$$

A common nonzero skew operator commuting with either triple would give a forbidden splitting by Lemma 3.2. Thus both common skew stabilizers vanish.

The norm bound. Consider one block of real dimension r , write its matrices as A_1, A_2, A_3 , and put $z_i = \dim \ker A_i$. For one matrix, a kernel of dimension z_i contributes $z_i(z_i - 1)/2$ to its skew stabilizer; a positive rotation level of multiplicity k contributes k^2 . The squared profile norm instead has contributions z_i^2 and $2k^2$. Hence twice the sum of the three stabilizer dimensions is $\|m\|^2 - \sum_i z_i$. The orbit sum map is submersive onto $\mathfrak{so}(r)$, because its orthogonal cokernel is the common skew stabilizer. Its kernel contains the simultaneous-conjugation tangent, of dimension $\dim \mathrm{SO}(r)$. Comparing dimensions gives

$$\|m\|^2 - \sum_i z_i \leq r^2 - r. \quad (4.4)$$

If the joint kernel is zero, the map

$$u \mapsto (A_1 u, A_2 u, A_3 u)$$

is injective into the kernel of addition from $\bigoplus_i \mathfrak{m} A_i$ to $\mathbb{R}^r$. This addition map is onto: the orthogonal complement of the sum of the images is the joint kernel. Its kernel has dimension $2r - \sum_i z_i$, so $\sum_i z_i \leq r$. Equation (4.4) yields $\|m\| \leq r$. The same proof works in odd dimension.

A joint kernel of dimension at least two would support a nonzero commuting skew operator. A possible joint kernel is therefore a single line, and it must belong to the block containing the sole possible zero of the moving weight. Remove this line. The reduced triple has neither a joint kernel nor a common skew stabilizer, so

$$\|m\|^2 = \|m_{\text{red}}\|^2 + 2 \sum_i z_{i,\text{red}} + 3 \leq (r-1)^2 + 2(r-1) + 3 = r^2 + 2.$$

The moving zero-level entry in the original profile has multiplicity two and pairs with zero in the other block's profile. Deleting that entry for the dot product improves the relevant squared bound to $r^2 - 2$. Cauchy–Schwarz is then strict. If there is a moving zero but no joint kernel, the bounds $\|m^P\| \leq r$, $\|m^Q\| \leq h$ are already strict in Cauchy–Schwarz, since the moving zero entry prevents proportionality. Thus $m^P \cdot m^Q < rh$ whenever the moving weight has a zero.

Excluding equality without a moving zero. The weak bound $m^P \cdot m^Q \leq rh$ holds by the same estimates. Equality forces proportional profiles, $\|m^P\| = r$, $\|m^Q\| = h$, and $\sum_i z_i = r$ in the first block, with the analogous equality in the second. The moving matrices are invertible. Each of the other two matrices has a kernel: if one were invertible, the other would vanish, and the remaining pair $A, -A$ would have $A \neq 0$ in its common skew stabilizer.

Both triples are absolutely irreducible. Indeed a proper common real invariant subspace has an invariant orthogonal complement. Each has even dimension, since the moving restriction is

invertible and skew. Together with the other original block this gives at least three orthogonal blocks, contrary to Lemma 3.2. A proper complex invariant subspace for the Hermitian triple gives a nonscalar commuting Hermitian projection $B + iD$, with B real symmetric and D real skew. Both commute with the real skew triple. A nonscalar B gives the forbidden real splitting, while nonzero D gives the forbidden skew stabilizer.

Lemma 4.1 now gives $r = h$ and identical profiles, including their numerical levels. Replace the second skew triple by an orthogonal copy of the first. Choose the isometry's orientation to match the moving special-orthogonal orbit. The other two matrices impose no additional orientation restriction: each has a kernel, and reflection in a kernel line is an orientation-reversing commuting isometry. Thus their orthogonal and special-orthogonal orbits coincide. On the sum of the two isometric copies, the skew operator rotating one copy into the other commutes with the triple. It gives a unitary, hence permutation-type, block in Lemma 3.2, again a contradiction. Equality is impossible. Equation (4.3) proves (4.2). $\square$

4.2 Core phases and the comparison function

The moment calculation has produced the strict numerical gap $R - p_\lambda - p_\mu - p_\xi > 0$. We now follow the cross-component comparisons along the schedule. Strict inversions alone do not record what happens on an interval where both positions and moving values agree. The phases defined next provide the correction for these double ties: such an interval will contribute -1 when its two phases are zero and $+1$ when they are one. The phase restriction is a consequence of the exceptional graph, not an arbitrary choice of a tie-breaking order.

For a signed position label a , define its *core phase* $p_a(t)$ on the open intervals between switch times. If its axis is off the unique cycle of its component, its phase is zero until the switch on the first edge toward that cycle and one afterward. If its axis lies on the cycle, its phase is $0, \frac{1}{2}, 1$, according as zero, one, or two of its cycle incidences have occurred. The phase is unchanged on replacing a by $-a$.

There is a useful algebraic description. Solve the component's root-basis equation with right-hand side e_a . The cumulative signed a -coordinate of the edges already used is $p_a(t)$. For an off-cycle source, pruning the attached trees leaves a unit contribution on its first edge toward the cycle. At a cycle source, the two incident cycle contributions are each one half. This proves the description. Since the actual coefficients are integral on bridges and half-odd-integral on cycle edges, phase $1/2$ also records that the translation term in the position formula has changed its fractional part by a half.

Lemma 4.3 (Plateaus). *Suppose cross labels a, c have equal position and equal assigned moving value on an open interval between switch times. Then*

$$(p_a, p_c) = (0, 0) \quad \text{or} \quad (1, 1). \quad (4.5)$$

Proof. The equality of moving values on distinct axes means that their assigned shape-coordinate forms agree on the whole minimal chamber face. Hence their shape values agree for every time. The initial coordinates have the same fractional part, and equality of the current positions implies that either both phases are half or neither is.

At any point of the plateau, reflect the entire suffix by the reflection exchanging the two signed axes, including every later position and switch root. This reflection fixes the position at the cut and fixes the current-frame shape $w\eta(t)$ for every t , since the two coordinate forms agree. Thus no new switch is needed at the cut, and the reflected endpoint remains in its original Weyl orbit. In

the root graph, cut each involved vertex into its past and future copies, then identify opposite-time copies crosswise. If both phases are half, each cut opens its cycle into a connected tree. The crosswise identification gives a connected l -edge graph, outside the exceptional configuration. If neither phase is half, one cut piece contains the cycle and the other is a tree, possibly an isolated vertex. The core piece uses the past copy at phase one and the future copy at phase zero. Opposite phases therefore join the two cores and leave a separate tree component. That too is excluded by Lemma 3.3. Only (4.5) remains. $\square$

For times between events, define

$$J(t) = \frac{1}{2} \sum_{a \in \hat{P}, c \in \hat{Q}} j_{a,c}(t), \quad j_{a,c}(t) = \begin{cases} 1, & (y_a - y_c)(v_a - v_c) < 0, \\ p_a + p_c - 1, & y_a = y_c \text{ and } v_a = v_c, \\ 0, & \text{otherwise.} \end{cases} \quad (4.6)$$

Use one-sided values at isolated equalities. The direct links of Lemma 3.7 give

$$J(0+) \geq R - p_\lambda - p_\mu, \quad J(1-) \leq p_\xi. \quad (4.7)$$

Indeed strict position and velocity orders must begin discordant and end concordant. An initial double tie has phases zero and contributes -1 , covered by subtracting both sorts of initial tie. At the final endpoint a double tie has phases one and contributes $+1$; a position tie with distinct velocities is an inversion just before the endpoint and is also counted by p_ξ . Away from the old switch times, any changing cross equality with distinct velocities gives the direct crossing link. Thus a decrease can occur only at an old switch time, which we call a *knot*.

Proposition 4.4 (Monotonicity at knots). *Under the standing no-integral-schedule assumption, the function J is nondecreasing at every knot, and hence on the whole interval in the one-sided sense used above.*

We prove this by reducing each knot to disjoint transfers of two moving values. A loss in their strict comparison count will either produce a local link or be compensated by a double-tie contribution.

4.3 The form of an exact knot

We now specify the shape genericity allowed by Lemma 3.4. This choice concerns the original exceptional root-basis schedule, which retains the limiting cover property; it imposes no cover condition on later augmented schedules. For fixed endpoint assignments and a fixed root basis, the endpoint equation determines every coefficient. The frames, coefficients, and translation parts therefore range over finitely many possibilities. Both switch conditions and equalities of signed positions consequently give a finite collection of affine level sets for shape values.

Let F be the span of the smallest chamber face containing μ . Its signed coordinate forms are $\pm M_j$, for independent variables M_j , with possible repetitions. No coordinate form is forced to vanish: a sole zero coordinate of μ is the free last coordinate in type D . On distinct axes, any equality at μ already identifies the corresponding forms on F .

Choose a polygonal shape, with slopes relatively interior in the face, that avoids simultaneous nonconstant affine level hits having nonproportional restrictions to F , and avoids such hits at its breaks. Such choices are dense: after subdivision, interior break values have open freedom in F , and a generic polygonal path avoids the finitely many affine sets of codimension at least two.

At any knot the nonzero velocity differences of its switches are therefore multiples of one form ℓ . Group signed position labels by their assigned forms modulo $\mathbb{R}\ell$. This partition is unchanged during the knot. If ℓ is proportional to $M_i \pm M_j$, $i \neq j$, its nontrivial classes are pairs of forms and their mirrors. If it is proportional to M_i , the only nontrivial class is $\{M_i, -M_i\}$. Consequently each class has either one value or two values $H > L$, ordered at μ . The sole possible exception, $M_i(\mu) = 0$ in an opposite class, contains only the two signs of one physical axis; it supports no root switch and may be omitted.

Across different classes, cross positions are unequal at the knot and cross velocity values are distinct. A swap changes their strict inversion count only if an external spectator lies strictly between both swapped positions and both swapped velocity values. The limiting cover property excludes exactly this situation by Corollary 2.14. Thus these comparisons are unchanged. Classes with one value need only the monotonicity of phases.

Lemma 4.5 (Matching at a knot). *Within each two-value class at a knot, the transfers of the high value form disjoint matching pairs inside each component, together with unused single labels. In an opposite class the matching is disjoint on physical axes, with its mirror transfers understood as the same physical switches.*

Proof. Every transfer sends H from a donor of lower position to a recipient of higher position. In an ordinary class, decompose this flow into directed paths from its net donors to its net recipients. Positions strictly increase, so no directed cycle occurs. Shortcut each path by its direct transfer. The resulting matching has the same final high and low assignments and legal positive gaps; see Equation (3.8). If the original transfers were not already disjoint pairs, this uses fewer switches, contrary to Lemma 3.3.

In an opposite class each physical swap flips two axes, producing two mirrored transfers. Mark each axis by its initial high sign. At any edge occurrence, the sum of the positions in the two *currently* high signs is strictly negative. At each vertex, pair consecutive incidences, leaving the first unpaired if their number is odd. The signed-position terms in each such pair cancel, because the high sign alternates. The paired incidences organize the edges into paths and cycles. A cycle would sum strict inequalities to $0 < 0$, and cannot occur. On a path the sum leaves just the positions of its two ends in their initial high signs, with negative sum. A direct flip of these two axes is therefore feasible. Distinct paths have disjoint unpaired ends; their flips realize exactly the original odd-incidence axes, hence exactly the same final sign assignment. Unless the old switches were already physically disjoint, this again reduces their number. $\square$

We may therefore compare units—single labels and directed pairs—of one component with units of the other. All local replacements below use distinct physical axes. Other switches at the knot are disjoint from these axes, apart from the automatic mirrors in an opposite class. A transfer $a \rightarrow b$ has root $e_b - e_a$; all transfers in the same two-value class have the same strictly increasing coefficient function, namely the assigned shape value of H minus that of L .

The replacements have exactly the same final assignment of H, L as the old switches. This preserves the whole shape, not merely its value at the knot: the old and new final frames differ on the right by an element of W_μ , which fixes F pointwise. Subsequent switches therefore have the same actual shape values and positions at the base parameter. The forgetting condition of Proposition 3.6 is satisfied. We will check its remaining hypotheses—lattice generation and feasible coefficient order and gaps—for each extra-edge replacement.

4.4 Pairs, single labels, and endpoint ties

The matching lemma leaves three kinds of comparison: two unused labels, a transfer and an unused label, and two transfers. We check that every decrease either produces a link or is offset by the contribution of a double tie.

Consider a transfer $a \rightarrow b$ at positions $x < z$, and a single label c in the other component at position h , initially high. The low case is obtained by taking signed mirrors. Omitting plateau terms, the strict count decreases by one precisely when

$$x \leq h \leq z. \quad (4.8)$$

Indeed before the knot only the comparison of c with the low recipient can contribute; afterward only its comparison with the new low donor can contribute. At an endpoint equality, the distinct approaching or departing velocities give respectively the inversion or concordance specified by the closed inequalities in (4.8).

Replace the transfer by

$$c \longrightarrow b, \quad a \longrightarrow c. \quad (4.9)$$

The final high value is at b , as required. The graph subdivides the old edge through the other component, with the same cycle signs; it is connected and unbalanced and generates the full root lattice. Its base coefficients are the common old coefficient, hence half-integral. Let d_1, d_2 be the derivatives of its two new coefficients under a primitive half-kernel direction. Their difference

$$D = d_2 - d_1$$

is nonzero: otherwise contracting the two new edges would give a dependency of the old basis. Choose its sign so that $D > 0$. The common increasing coefficient function then orders the two new times correctly. If $x < h < z$, both gaps are strictly positive and (4.9) is a forbidden feasible link.

It remains to compute the derivative of a zero gap. For an ordered switch root β_i , use Equation (3.3). The two new roots are $\beta_1 = e_b - e_c$, $\beta_2 = e_c - e_a$, with $\beta_1 \cdot \beta_2 = -1$.

A tie at the donor. If $h = x$, the second new gap is zero. Since the old root is $\beta_1 + \beta_2$, write

$$d_1\beta_1 + d_2\beta_2 = d_1(\beta_1 + \beta_2) + D\beta_2.$$

The full derivative has zero root sum. Absorbing d_1 into the old edge therefore gives an old-basis equation with right-hand side $-D\beta_2 = D(e_a - e_c)$. By the algebraic description of phases, its pre-knot contributions at a, c are $Dp_a^-, -Dp_c$. Differentiating (3.3), including the first new switch, gives

$$G'_2 = D(p_a^- + p_c) + d_1 - d_2 = D(p_a^- + p_c - 1). \quad (4.10)$$

Before the knot, a, c form a double plateau. If both phases are one, (4.10) is positive and gives a feasible link. Otherwise Lemma 4.3 makes both phases zero. Losing the plateau contribution -1 then compensates for the strict-count loss of one.

A tie at the recipient. If $h = z$, the first new gap is zero. This time write the new derivative as $d_2(\beta_1 + \beta_2) - D\beta_1$. Absorbing d_2 into the old edge gives right-hand side $D\beta_1 = D(e_b - e_c)$ in the old-basis equation. At the old edge, its contribution to the b -coordinate is d_2 , so $p_b^+ = p_b^- + d_2/D$. Equation (3.3) gives

$$G'_1 = -D(p_b^- + p_c) - d_1 = D(1 - p_b^+ - p_c). \quad (4.11)$$

The ensuing plateau of b, c has either two zero phases, giving a feasible link, or two unit phases. In the latter case its new contribution $+1$ compensates for the strict loss. Thus every pair-single

comparison is nondecreasing when links are excluded. Two singles can decrease only by a crossing with distinct velocities. Their axes are unused by the knot, so this is a direct link by Lemma 3.7.

4.5 Two pairs and the star replacement

Consider $a \rightarrow b$ at positions $x < z$ and $c \rightarrow d$ at positions $h < k$. Before the knot the two mixed comparisons count when $x \leq k$ and $h \leq z$; after it they count when $x > k$ and $h > z$. Thus, without plateau terms, the loss is two precisely when the closed intervals $[x, z]$, $[h, k]$ overlap, and zero otherwise. The endpoint convention again follows from one-sided times. For example, if $x = k$, the high donor a lies just below the low recipient d before the knot, giving an inversion; after the knot the same position order is concordant with their exchanged values.

A touching mixed endpoint permits the switched matching

$$a \longrightarrow d, \quad c \longrightarrow b$$

with a zero gap and the correct other weak signs. Deleting that zero-gap switch leaves fewer than l edges. Suppose henceforth that the overlap is strict. The two old coefficients are equal. If they are half-odd-integral, both edges lie on their components' cycles. Switching the matching joins the two opened cycles into a connected l -edge graph, again outside the exceptional configuration. It is immaterial whether the resulting connected graph is balanced: a balanced graph gives the permutation reduction.

The remaining case has integral common coefficient. Both old edges are bridges. For each bridge, say that its *core is at an endpoint* if the piece containing that endpoint after removal of the bridge contains the cycle. This switch advances the phase from zero to one at the endpoint away from the core and leaves the core-side endpoint phase unchanged.

Suppose the donor positions differ, say $x < h$. Use the star at c : send c to one of b, d , then a to c , then c to the other of b, d . Strict overlap implies that all three gaps are positive. The new graph joins the four pieces of the two removed bridges by this star. It is connected, retains both unbalanced cycles, and generates the full root lattice. Along the path between its two cores, the primitive half-kernel coefficients have magnitude one; off that path the connecting-tree coefficients vanish. In the displayed root orientations, their possibilities are as follows, up to reversing their common sign:

Core endpoints	$(d_{c \rightarrow b}, d_{a \rightarrow c}, d_{c \rightarrow d})$	Wing order feasible?
a, c	$(0, 1, 0)$	no
a, d	$(0, 1, 1)$	yes
b, c	$(1, 0, 0)$	yes
b, d	$(1, 0, -1)$	yes

To see the signs directly, only the star edges on the path from the first core endpoint to the second are active. Conservation at c gives equal coefficients on $a \rightarrow c, c \rightarrow d$, and opposite coefficients on $c \rightarrow b, c \rightarrow d$. This gives exactly the table. All three coefficients use the same increasing shape-level function. One can therefore order the wings so that their derivatives surround the middle derivative except when both cores are at donors. Equal derivatives give equal times, with the stated switch order at that time. In every other case this is a feasible link.

For a receiver star, if $z < k$, use

$$a \longrightarrow b, \quad b \longrightarrow d, \quad c \longrightarrow b,$$

allowing the two wing transfers into b to be interchanged. Strict overlap makes all gaps positive. The wing derivatives are those of $a \rightarrow b$ and $c \rightarrow b$; they must bracket the middle derivative on $b \rightarrow d$. If the cores are at b, d , only the middle edge lies on the path between them, so its nonzero derivative cannot be bracketed by the two zero wing derivatives. In every other case, the active path uses the two wings with opposite derivatives, the middle and one wing with equal derivatives, or one wing alone with the other two derivatives zero. Choosing the common sign and the wing order makes the times weakly increasing. Thus only the case of two recipient cores is infeasible. If $k < z$, interchange the two original pairs.

If no link exists, the donor-star and receiver-star obstructions cannot both hold. Thus $x = h$ or $z = k$, or both. These same-role coincidences produce double plateaus on both sides of the knot. If only the donors coincide, the receiver star forces both cores to be at recipients. Each donor phase then jumps from zero to one, so their plateau contribution rises from -1 to $+1$. This compensates for the strict-count loss of two. If only the recipients coincide, the donor star forces both cores to be at donors, and the same compensation occurs at the recipients. If both pairs of endpoints coincide, the sum of the two plateau increments is two by the bridge-phase rule, again the required compensation. This exhausts all pair-pair comparisons. Mirrored transfers give the same calculations; the factor $1/2$ in (4.6) applies uniformly.

Proof of Proposition 4.4. Comparisons across distinct form classes are unchanged by the cover property. Within a class, Lemma 4.5 reduces all comparisons to the single-single, pair-single, and pair-pair cases just proved. Each possible loss either gives a feasible extra-edge link, forbidden by Proposition 3.6, or is compensated by its plateau contribution. The direct crossing links exclude decreases away from knots. Summing gives the assertion. $\square$

Completion of the proof of Theorem 3.1. Proposition 4.2 and (4.7) imply $J(0+) > J(1-)$, contradicting Proposition 4.4. This proves integral realizability for the generic shapes used above. The uniform bounded-switch closedness of Lemma 3.4 passes to an arbitrary shape with the same endpoint. Together with the permutation case and the rank reduction of Lemma 3.3, this completes that theorem. $\square$

5 From integral schedules to tensor intertwiners

The switch integrality in Definition 2.1 is weaker than the usual integral saturated-chain condition in a highest-weight path model. We now prove the representation-theoretic implication that is needed here directly, using fundamental columns.

Proposition 5.1 (Column intertwiners). *Let λ, μ, ξ be dominant integral weights of type D_n , $n \geq 2$. Choose a shape obtained by concatenating straight segments whose increments are fundamental weights and whose sum is μ . If this shape admits an integral schedule from λ to $W\xi$, then*

$$\mathrm{Hom}_G(V(\lambda) \otimes V(\mu), V(\xi)) \neq 0.$$

An empty list of columns is allowed when $\mu = 0$.

We first record carefully what happens to integrality when a schedule is folded into the dominant chamber.

Lemma 5.2. *An integral schedule can be folded into a dominant path with Weyl direction states in decreasing Bruhat order. Its changes of state can be expressed as an ordered sequence of downward reflections, including at coincident times. At each such step, from w to $s_\alpha w$ at time t ,*

$$\alpha \cdot w\eta(t) \in \mathbb{Z}. \quad (5.1)$$

For a concatenation of fundamental columns, all column endpoint positions of the folded path are dominant integral weights.

Proof. Use the simplifications and folding construction in the proof of Lemma 2.6. After deleting removable switches, perturb the remaining strict schedule to separate its sites and make ordinary chamber crossings transverse. In the perturbed path, an original switch becomes a downward reflection in the current chamber frame, and a sorting crossing becomes a downward simple reflection. There are at most the longest Weyl length many such strict downward steps. Taking a subsequence fixes their discrete labels and order; their times then converge. This gives the asserted ordered chain even if several steps acquire the same limiting time. The perturbations used for this construction need not be integral.

Integrality is checked on the limiting chain, in its specified order. For an original switch, changing the chamber frame changes both its root and its direction by the same Weyl element. Its pairing in (5.1) is therefore the original integral pairing. For a sorting crossing, suppose all preceding coefficients are integral and write the position before that crossing as

$$x = \lambda + w\eta - \sum_{j < i} m_j \beta_j. \quad (5.2)$$

The wall equation $\alpha \cdot x = 0$ gives

$$\alpha \cdot w\eta = -\alpha \cdot \lambda + \sum_{j < i} m_j \alpha \cdot \beta_j \in \mathbb{Z}.$$

The usual switch identity preserves (5.2) after the crossing. Induction therefore proves integrality for every prefix, also inside a limiting cluster. At a column endpoint η is a weight, so (5.2) shows that x is a weight. It is dominant by construction. $\square$

Consider a single column of weight ω . Normalize its parameter to $0 \leq \tau \leq 1$, so that the original shape on this interval is $\chi + \tau\omega$, with χ a sum of preceding fundamental weights. Write x_0, x_1 for its dominant endpoints and put $\delta = x_1 - x_0$. Let u and v be the first and last Weyl states in its open interior. Boundary-time steps are left between the columns; thus the last state of one column dominates the first state of the next. Since $\alpha \cdot w\chi$ is integral, Lemma 5.2 gives at every interior step

$$\tau \alpha \cdot w\omega \in \mathbb{Z}. \quad (5.3)$$

Minuscule columns. For a minuscule weight, all root pairings are 0 or ± 1 . Consequently (5.3) permits no interior reflection that changes $w\omega$. The column is straight, with $\delta = u\omega$. The extremal case of Lemma 2.3 gives an intertwiner

$$V(x_0) \otimes V(\omega) \longrightarrow V(x_1)$$

whose highest-to-highest coefficient is nonzero at the fixed flag u . This includes the vector column and both spin columns. For D_2 and D_3 all fundamental columns are minuscule. The classical

exterior and half-spin models are described in [10, Lectures 19–20]; here only their root-pairing property is needed for the minuscule case. Although full Weyl states inside the column may change while fixing ω , this causes no problem for composition: the chosen fixed waypoint has future u , and $u \geq v$ dominates the next column's initial state.

It remains to treat an exterior column

$$\omega = e_1 + \cdots + e_k, \quad 2 \leq k \leq n-2.$$

Let E be the standard $2n$ -dimensional quadratic space, with basis E_{+i}, E_{-i} satisfying $(E_{+i}, E_{-j}) = \delta_{ij}$ and all other basis pairings zero. The module $\bigwedge^k E$ is the fundamental module $V(\omega)$; its flag vectors are wedges of isotropic k -planes.

The coefficient to be constructed. For the exterior column, Lemma 2.3 reduces the intertwiner problem to a functional supported at weight δ . The following criterion isolates its annihilation conditions. We will then construct such a functional and a flag at which it is nonzero, with past and future limits u, v so that columns can be composed.

Lemma 5.3. *Let J be the set of simple roots α_i with $x_0 \cdot \alpha_i = x_1 \cdot \alpha_i = 0$. A functional supported on the weight space of weight δ in $\bigwedge^k E$, invariant under the semisimple group generated by J , is an allowed highest-to-highest coefficient for an intertwiner $V(x_0) \otimes V(\omega) \rightarrow V(x_1)$.*

Proof. Put $h_i = x_0 \cdot \alpha_i$ and $h'_i = x_1 \cdot \alpha_i$. By Lemma 2.3, the functional must annihilate $f_i^{h_i+1}V(\omega)$ for every i . A vector contributing to its weight- δ component would have source weight $\delta + (h_i + 1)\alpha_i$, whose simple-root pairing is

$$(\delta + (h_i + 1)\alpha_i) \cdot \alpha_i = h_i + h'_i + 2.$$

All weights of $\bigwedge^k E$ have coordinates in $\{0, 1, -1\}$, so this pairing cannot exceed two. Since h_i, h'_i are nonnegative integers, the source is absent unless both vanish. In that case $\alpha_i \in J$, and invariance annihilates $f_i V(\omega)$. $\square$

The midpoint and a subgroup waypoint. The nonzero root pairings of ω have absolute value at most two. By (5.3), an effective interior switch occurs only at $\tau = \frac{1}{2}$ and has absolute pairing two. It flips the signs on two occupied axes. A root with pairing zero preserves both the occupied support and its complement. Thus the support S of $w\omega$ is fixed throughout the column, every interior reflection belongs to $D_S \times D_{S^c}$, and

$$a = u\omega, \quad a' = v\omega, \quad \delta = \frac{a + a'}{2}. \quad (5.4)$$

In particular, the set

$$R = \{i \in S : a'_i = -a_i\}$$

has even cardinality. On $S \setminus R$ one has $\delta_i = a_i = a'_i \in \{1, -1\}$, whereas $\delta_i = 0$ on R .

Let H be the connected subgroup for $D_S \times D_{S^c}$. The orbit $H \cdot uB$ is a flag variety of H : its stabilizer at uB is the Borel $H \cap uBu^{-1}$, and the ambient positive Borel induces the positive Borel $H \cap B$. Each interior comparison is a downward Bruhat reflection in this orbit. Indeed its root is positive for $H \cap B$, and its inverse image by the current state is negative for the corresponding isotropy Borel, exactly as in the ambient comparison. The resulting Bruhat inequality supplies a waypoint

$$b \in U_H^+ u \cap U_H^- v \subset U^+ u \cap U^- v. \quad (5.5)$$

The two unipotent groups here are the positive and negative ones induced from the ambient group. Equivalently, one may obtain this intersection by the Bruhat bridge construction used in Lemma 2.4.

The wedge at b represents a maximal isotropic plane in $E_S = \text{span}\{E_{+i}, E_{-i} : i \in S\}$. Its a and a' coordinates are nonzero, since their lines are its past and future limits. Use the a coordinate as a graph chart: write its basis as

$$z_j = E_{a_j j} + \sum_{i \in S} T_{ij} E_{-a_i i} \quad (j \in S).$$

Here the notation $E_{a_i i}$ denotes E_{+i} or E_{-i} according to the sign a_i . Isotropy says that T is skew-symmetric. The a' coordinate of $\bigwedge_{j \in S} z_j$ is, up to sign, $\det T_{R,R}$, so

$$T_{R,R} \text{ is nonsingular.} \quad (5.6)$$

When R is empty, the empty determinant is understood to be one.

There is a useful restriction on its entries:

$$i < j, \quad T_{ij} \neq 0 \quad \implies \quad a_i = -1. \quad (5.7)$$

Indeed a single replacement in the wedge gives a nonzero coordinate of weight $a - a_i e_i - a_j e_j$. Since a is the past extremal limit in (5.5), the difference $-a_i e_i - a_j e_j$ has positive pairing with the regular dominant flow direction. This difference is a root, so it is positive. In type D , a root supported on $i < j$ is positive precisely when its i -coefficient is positive, which proves (5.7).

A common-wall invariant functional. We now have a waypoint with the required flow limits and a nonsingular principal matrix $T_{R,R}$. We first construct a weight- δ functional satisfying Lemma 5.3. The free coefficients in this construction will then be chosen to make its value at the waypoint nonzero.

The sign change on the last coordinate preserves the positive root system, interchanging its two end simple roots. Apply it, if needed, to arrange $x_0 \geq 0$ coordinatewise; this changes none of the preceding arguments. Let $\mathcal{B}$ be the collection of blocks of axes having $\delta_i = 0$ and the same positive value of x_{0i} . Assign a scalar c_B to each such block, put $c_i = c_B$ for $i \in B$, and put $c_i = 0$ on the zero-position, zero- δ tail. Define

$$\Theta = \sum_{B \in \mathcal{B}} c_B \sum_{i \in B} E_{+i}^* \wedge E_{-i}^*, \quad \Psi = \left(\bigwedge_{i: \delta_i \neq 0} E_{\delta_i i}^* \right) \wedge \Theta^{|R|/2}. \quad (5.8)$$

Choose any fixed order for the coordinate wedge. The functional Ψ has degree k and is supported on the weight δ .

It is invariant under the common-wall semisimple group. To see this directly, a common wall $e_i - e_{i+1}$ requires equal x_0 coordinates and equal δ coordinates. On a block with nonzero δ , the coordinate wedge in (5.8) is the determinant covector and is invariant under the special-linear group. On a block with positive x_0 and zero δ , the two-form $\sum E_{+i}^* \wedge E_{-i}^*$ is the canonical invariant pairing form for the standard space and its dual, hence is invariant under the whole general-linear group. The remaining zero-position, zero- δ tail carries a special-orthogonal factor and contributes no covectors. If the other end root $e_{n-1} + e_n$ belongs to J , then $x_{0,n-1} = x_{0,n} = 0$. Dominance of x_1 gives $(\delta_{n-1}, \delta_n) = (0, 0)$ or $(1, -1)$. The first case belongs to the orthogonal zero tail. In the second, the difference end root is absent, and changing the last sign gives a special-linear determinant block. These cases cover the group generated by J .

Nonzero evaluation at the waypoint. It remains to choose the c_B so that Ψ is nonzero at b . The coordinate factors in (5.8) select exactly their own primal columns in the graph chart. After these factors are removed, the pullback of Θ to the remaining columns indexed by R has skew matrix

$$\tilde{T}_{ij} = (a_i c_i + a_j c_j) T_{ij}, \quad i, j \in R, \quad (5.9)$$

up to one overall sign convention. If ζ_i are the coordinate functions on the primal plane, then $E_{+i}^* \wedge E_{-i}^*$ pulls back to $a_i d\zeta_i \wedge \sum_j T_{ij} d\zeta_j$; summing the two contributions to an entry gives (5.9). The top exterior power is nonzero precisely when $\tilde{T}$ is nonsingular. This is automatic for $R = \emptyset$; assume henceforth that $R \neq \emptyset$.

Partition $R = N \sqcup P$, where

$$N = \{i \in R : x_{0i} > 0, a_i = -1\}.$$

Thus P consists of positive signs at positive positions, together with all indices in the zero tail. Dominance of the first part of the column gives dominance of $x_0 + \varepsilon a$ for all sufficiently small $\varepsilon > 0$. Within an equal positive-position block, the signs therefore occur in decreasing order. In the zero tail no negative sign can precede another index: only the final coordinate may be negative. Combined with (5.7), this proves

$$T_{R,R} = \begin{pmatrix} A & B_0 \\ -B_0^\top & 0 \end{pmatrix} \quad \text{in the ordering } N, P. \quad (5.10)$$

Furthermore a nonzero entry from $i \in N$ to a positive-position $j \in P$ must have $i < j$ and $x_{0i} > x_{0j}$. Equality of the positions would contradict the decreasing sign order in that block. For a zero-tail $j \in P$, every $i \in N$ already precedes j , regardless of the sign of a possible final coordinate.

Order the blocks in $\mathcal{B}$ from larger to smaller position. Choose positive exponents s_B strictly decreasing in this order, write B_i for the block containing a positive-position index $i \in R$, and set

$$c_B = 1 + t^{s_B} \quad (t > 0).$$

We let $t \rightarrow 0^+$. For a permitted N, P entry with both positions positive, the larger-position block has the larger exponent, so its perturbation decays faster. This separates the blockwise coefficients while leaving their common limit equal to one. In (5.9), the N, N block tends to $-2A$. For a positive-position P index $j \in B$, scale its row and column by t^{-s_B} . On a permitted entry from $i \in N$ to this j , the multiplier becomes

$$t^{-s_B}(-c_i + c_j) = 1 - t^{s_{B_i} - s_B} \rightarrow 1.$$

For a zero-tail P index leave the row and column unscaled and then negate both: its unscaled multiplier is $-c_i \rightarrow -1$. The P, P block stays zero. Thus the scaled modified matrix tends to

$$\begin{pmatrix} -2A & B_0 \\ -B_0^\top & 0 \end{pmatrix}. \quad (5.11)$$

This matrix is nonsingular. More generally, if $\gamma \neq 0$ and (x, y) lies in the kernel after replacing A in (5.10) by γA , then $(x, y/\gamma)$ lies in the kernel of the original matrix. Nonsingularity therefore follows from (5.6). By continuity, $\tilde{T}$ is nonsingular for sufficiently small positive t . This proves that the functional Ψ evaluates nontrivially on the waypoint wedge. Lemma 5.3 now supplies the required single-column intertwiner.

Proof of Proposition 5.1. Apply Lemma 5.2 and perform the preceding construction on each column. Write its endpoints as h_{j-1}, h_j , its weight as ω_j , and choose the resulting intertwiner

$$V(h_{j-1}) \otimes V(\omega_j) \longrightarrow V(h_j).$$

Its highest-to-highest coefficient is nonzero on the chosen waypoint. For an exterior column the past and future flags are its first and last states u_j, v_j ; for a minuscule column both may be taken to be u_j . At a column boundary, $v_j \geq u_{j+1}$. Thus the chosen future flag dominates the next past flag: directly for an exterior column, and through $u_j \geq v_j \geq u_{j+1}$ for a minuscule column. Lemma 2.4 therefore makes the composition nonzero on the Cartan summand

$$V\left(\sum_j \omega_j\right) = V(\mu) \subset \bigotimes_j V(\omega_j).$$

The first and last dominant positions are $h_0 = \lambda$ and $h_q = \xi$, proving the assertion. For an empty list the folded path is constant, so $\lambda = \xi$ and the identity map suffices. $\square$

6 Removing a zero tail and completing the proof

Theorem 3.1 applies when the moving weight has at most one zero coordinate. We now supply the real existence input needed when all three labels have a longer zero tail. The input is a tensor occurrence for labels supported on a common initial coordinate segment. We construct an occurrence for twice the truncated labels in a smaller even spin group. Passing to real moments then removes the factor of two. The integral schedule will be constructed for the unscaled truncated labels, using the restriction of the original column shape.

Write the coordinates of the labels as $\lambda = (L_1, \dots, L_n)$, $\mu = (M_1, \dots, M_n)$, and $\xi = (X_1, \dots, X_n)$. The reduction we shall prove is the following.

Proposition 6.1 (Removal of a zero tail). *Suppose λ, μ, ξ are dominant integral D_n -weights, all supported in coordinates $1, \dots, r$, where $1 \leq r \leq n - 2$. If $V(\xi)$ occurs in $V(\lambda) \otimes V(\mu)$, then*

$$V(2\hat{\xi}) \text{ occurs in } V(2\hat{\lambda}) \otimes V(2\hat{\mu}) \text{ for } D_{r+1},$$

where $\hat{\lambda} = (L_1, \dots, L_r, 0)$, and similarly for the other two labels.

6.1 A preprojective-module criterion

Lusztig's semicanonical basis relates enveloping algebras to preprojective representation varieties [29]. We use its generic-component coordinates, together with their left and right filtrations [12, 8], to turn tensor occurrence into simultaneous bounds on the top and socle of one module. Those bounds will let us change the two orthogonal leaves while controlling both tensor factors.

Let $\mathcal{D}$ be a finite simply laced Dynkin diagram, possibly disconnected. Choose an orientation of its edges and adjoin a reverse arrow to every arrow. A representation of the resulting doubled quiver is *preprojective* if, at each vertex, the signed sum of the two-arrow return maps is zero. Our sign convention assigns uv a plus sign at j and vu a minus sign at i , for an oriented arrow $u : i \rightarrow j$ with reverse v . Write $\Lambda(d)$ for the variety of these representations on fixed spaces of dimension vector $d = (d_i)$.

For $M \in \Lambda(d)$, set

$$\text{top}_i M = \dim \left(M_i / \sum_{a: j \rightarrow i} \text{im } M_a \right), \quad \text{soc}_i M = \dim \bigcap_{a: i \rightarrow j} \ker M_a.$$

Both sums and intersections run over arrows of the doubled quiver.

Proposition 6.2 (Tensor criterion). *Let λ, μ, ξ be dominant integral weights for $\mathcal{D}$, and suppose $\lambda + \mu - \xi = \sum_i d_i \alpha_i$ with $d_i \in \mathbb{Z}_{\geq 0}$. Put $\lambda_i = \lambda \cdot \alpha_i$ and $\mu_i = \mu \cdot \alpha_i$. Then*

$$\begin{aligned} \text{Hom}(V(\lambda) \otimes V(\mu), V(\xi)) &\neq 0 \\ \iff \exists M \in \Lambda(d) : \quad \text{top}_i M &\leq \lambda_i, \quad \text{soc}_i M \leq \mu_i \quad \text{for every } i. \end{aligned} \quad (6.1)$$

Proof. We recall precisely the semicanonical facts used here. On the preprojective varieties, convolution of invariant constructible functions is defined by integration with compactly supported Euler characteristic over submodules, with the submodule as the right factor. The constant function on the unique representation of dimension e_i represents f_i . This identifies $U = U(\mathfrak{n}^-)$ with the algebra generated by these functions; we identify the positive enveloping algebra in the cited conventions with the negative one by the Chevalley isomorphism. In each degree, generic evaluation on irreducible components is a vector-space isomorphism

$$U_d \longrightarrow \mathbb{C}^{\text{Irr } \Lambda(d)}. \quad (6.2)$$

Moreover, the image of $(f_i^p U)_d$ is the coordinate subspace supported on components whose generic i -top is at least p . The image of $(U f_i^p)_d$ is the analogous subspace for generic i -socle. Finite Dynkin type needs no additional nilpotence condition. For the constructible-function realization and generic component coordinates, see [12, Proposition 3.1 and Sections 5.2–5.6]; for their compatibility with the top and socle filtrations, see [8, Lemmas 5.1, 5.4, 5.6, and 5.9].

Here is the geometry behind the filtration statement, which also fixes the left/right convention. The preprojective equations describe the union of conormals to the finitely many ordinary-quiver orbits. Its components have dimension

$$D(d) = \sum_{\{i,j\} \in \mathcal{D}} d_i d_j$$

and are counted by the Kostant partitions of d , hence by $\dim U_d$. On the stratum with i -top exactly p , remove the top by replacing M_i with its incoming image, of dimension $a = d_i - p$. The lower representation has i -top zero. Conversely, put $S = \sum_{j \sim i} d_j$. After choosing the retained a -plane, extensions add outgoing maps on the remaining p dimensions with values in the kernel of the signed incoming map, of dimension $S - a$. Their parameter count, including the Grassmannian, is

$$p(S - a) + pa = pS = D(d) - D(d - pe_i).$$

The other relations are unchanged, because a path entering vertex i enters its retained image. This bundle construction gives the component correspondence for exact top p .

At exact top p , the submodule with quotient S_i^p is unique, so multiplication by $f_i^{(p)} = f_i^p/p!$ transfers the generic value from the lower component. It vanishes when the top is smaller than p . Descending induction on the top removes any contributions on larger-top components. Together with induction on dimension and the component count, this explains both (6.2) and its left filtration. Transposition of the arrows gives the right filtration and socle. Thus the same generic-coordinate basis is compatible with all the subspaces just described.

By Lemma 2.3 and the highest-generator presentation of $V(\mu)$, the multiplicity in question is the dimension of

$$U_d / \left(\sum_i f_i^{\lambda_i+1} U + \sum_i U f_i^{\mu_i+1} \right)_d. \quad (6.3)$$

Under (6.2), this quotient is nonzero exactly when there is a component on which all the generic top and socle bounds in (6.1) hold. Those bounds are open rank conditions. Their intersection on such a component is nonempty; conversely, any point satisfying them lies on a component where they hold generically. This proves the criterion. $\square$

6.2 Bounds along the chain

To shorten the diagram, we will retain an initial chain and replace its continuation by two leaves. We first bound the incoming-image annihilator and outgoing kernel at each possible cut, on every level of the nilpotent filtration.

For D_n , use simple roots $\alpha_i = e_i - e_{i+1}$ for $i < n$, and $\alpha_n = e_{n-1} + e_n$. Suppose that the occurrence in Proposition 6.2 holds, and choose a module from that criterion. Write $p = n - 1$. Combine the two leaf spaces into

$$V_p = W = W_+ \oplus W_-,$$

following the trunk $V_1, \dots, V_{p-1}$, and put $V_0 = 0$. With arrows oriented to the right, denote the paired maps on edge i by

$$u_i : V_{i-1} \longrightarrow V_i, \quad v_i : V_i \longrightarrow V_{i-1}.$$

Here u_p combines the two leaf maps and v_p combines their reverses. The relations are

$$t_i = u_i v_i = v_{i+1} u_{i+1} \quad (i < p), \quad t_p = u_p v_p, \quad (t_p)_{++} = (t_p)_{--} = 0. \quad (6.4)$$

Since $t_1 = 0$, all t_i are nilpotent. This follows inductively from the fact that uv is nilpotent if and only if vu is nilpotent. With $\Delta_j = L_j + M_j - X_j$, the dimensions are

$$\dim V_i = \sum_{j \leq i} \Delta_j \quad (1 \leq i \leq p). \quad (6.5)$$

The last equality gives the sum of the two original leaf dimensions. Top bounds on the trunk are $L_i - L_{i+1}$, and the two leaf bounds are $L_p - L_n, L_p + L_n$; socle bounds use M in place of L .

For a single edge, the nilpotent operator

$$\begin{pmatrix} 0 & v_i \\ u_i & 0 \end{pmatrix} \quad \text{on } V_{i-1} \oplus V_i$$

has a Jordan basis whose rows alternate between the two spaces. Such a basis is obtained by choosing homogeneous top generators of each Jordan length. In one row, let h_L, h_R be the numbers of vectors on the left and right, and let $T, S \in \{0, 1\}$ record whether its top and bottom, respectively, lie on the right. Then

$$h_R = h_L - 1 + T + S. \quad (6.6)$$

For $b \geq 1$, define

$$q_i(b) = \dim(\ker t_i \cap \operatorname{im} t_i^{b-1}), \quad q_0(b) = 0,$$

the number of t_i -rows of length at least b . Let $K_i^T(b)$ be the sum of T over edge- i rows with $h_R \geq b$, and let $K_i^T(b)$ use $h_L \geq b$. Define $K_i^S, K_i'^S$ similarly. These quantities do not depend on a common choice of bases for adjacent edges.

Lemma 6.3 (Threshold bounds). *For every $1 \leq r \leq p$ and $b \geq 1$,*

$$K_r^T(b) \leq L_r + \frac{q_r(b)}{2}, \quad K_r^S(b) \leq M_r + \frac{q_r(b)}{2}. \quad (6.7)$$

Proof. Fix b and suppress it from the notation. For each vertex, write

$$F_i^*(b) = \ker t_i^* \cap \operatorname{im}(t_i^*)^{b-1}$$

for its dual high-socle space. At a trunk vertex $i < p$, the two incoming maps give

$$\begin{aligned} \dim(\operatorname{Ann}(\operatorname{im} u_i) \cap F_i^*(b)) &= K_i^T, \\ \dim(\operatorname{Ann}(\operatorname{im} v_{i+1}) \cap F_i^*(b)) &= q_i - K_{i+1}^{T'}. \end{aligned}$$

These identities are read directly from the alternating rows: the left image fails to contain the right-hand top exactly when $T = 1$, whereas the right image fails to contain a left-hand top exactly when $T = 0$. The intersection of the two annihilators is bounded by the full vertex top. The dimension inequality therefore gives

$$K_i^T \leq L_i - L_{i+1} + K_{i+1}^{T'} \quad (i < p). \quad (6.8)$$

At W , the space $F_p^*(b)$ splits into its two colors, since t_p is off diagonal. Let their dimensions be q_+, q_- , so $q_+ + q_- = q_p$. The left incoming-image annihilator has intersection dimension K_p^T with $F_p^*(b)$. Its intersection with either color is bounded by the corresponding leaf top, whence

$$K_p^T - q_- \leq L_p + L_n, \quad K_p^T - q_+ \leq L_p - L_n$$

after naming the colors accordingly. Adding proves

$$K_p^T - \frac{q_p}{2} \leq L_p. \quad (6.9)$$

No equality of the old color dimensions is required.

For each edge, (6.6) also gives

$$K_i^{T'} - \frac{q_{i-1}}{2} \leq K_i^T - \frac{q_i}{2}. \quad (6.10)$$

Indeed, the contribution of one row to right minus left is

$$\left(\mathbf{1}_{\{h_R \geq b\}} - \mathbf{1}_{\{h_L \geq b\}} \right) \left(T - \frac{1}{2} \right).$$

It is zero if the indicators agree. If a row is gained on the right, then $T = S = 1$; if it is lost, then $T = S = 0$. In either case its contribution is $1/2$.

Combining (6.8) with (6.10) for edge $i + 1$ gives

$$K_i^T - \frac{q_i}{2} \leq L_i - L_{i+1} + K_{i+1}^T - \frac{q_{i+1}}{2}.$$

Starting from (6.9) and descending proves the top assertion. At a trunk vertex, on the primal high-socle space $\ker t_i \cap \operatorname{im} t_i^{b-1}$, the outgoing kernels $\ker v_i$ and $\ker u_{i+1}$ have intersection dimensions K_i^S and $q_i - K_{i+1}^{TS}$, respectively. Their common intersection is bounded by the vertex socle. The same leaf estimate and row comparison, with T, L replaced by S, M , therefore prove the socle assertion. $\square$

6.3 Positioning subspaces in a nilpotent filtration

At each possible cut, the threshold estimates now control the incoming annihilator and outgoing kernel on every nilpotent filtration level. We next determine how such profiles bound intersections with a moving subspace, and which motions can be induced by automorphisms commuting with the nilpotent operator.

Lemma 6.4 (Generic filtered intersection). *Let*

$$H = F_1 \supseteq F_2 \supseteq \cdots \supseteq F_{s+1} = 0$$

be a filtration of a finite-dimensional complex vector space, and let $P \subseteq \mathrm{GL}(H)$ be its stabilizer. For subspaces A, C write $a_b = \dim(A \cap F_b)$, $c_b = \dim(C \cap F_b)$, and $f_b = \dim F_b$. For g in a nonempty Zariski-open subset of P ,

$$\dim(A \cap gC) = \max \left(0, \max_{1 \leq b \leq s} (a_b + c_b - f_b) \right). \quad (6.11)$$

Proof. Each term on the right is a lower bound by the dimension inequality inside F_b . To attain all the bounds, choose coordinates adapted to the filtration. A basis for a subspace with profile a_b consists of $a_b - a_{b+1}$ generic vectors with allowed coordinates in F_b , for each b ; the analogous description applies to C . Such bases have the required profiles on a nonempty open set.

The coordinate supports of the combined list of basis vectors are nested. This is the nested-support case of Hall's distinct-representatives criterion [14, Theorem 1]. A maximum matching of these vectors to distinct allowed coordinates leaves exactly

$$\max \left(0, \max_b (a_b + c_b - f_b) \right)$$

unmatched vectors. Indeed any collection of vectors has, as its union of allowed coordinates, its largest allowed filtration level. For level F_b , at most $a_b + c_b$ vectors are constrained to that level or deeper. These are all the matching constraints, and they are attained by matching from the deepest level outward. A matching gives a nonzero minor, so the generic rank of the combined list has exactly this deficiency. Since each separate list is independent, the deficiency is its intersection dimension.

Finally, P is transitive on subspaces with a prescribed filtration profile: extend a filtration-adapted basis of the subspace to one of H . We may therefore hold A fixed and vary only C , which gives (6.11). Maximal rank is an open condition. $\square$

We next recall the centralizer's action on the socle filtration. This is dual to the quotient-filtration statement in [9, Section 3.6]; we include the short Jordan-basis proof.

Lemma 6.5 (Centralizer on the socle). *Let t be nilpotent on a complex vector space V , and put*

$$F_b(t) = \ker t \cap \mathrm{im} t^{b-1} \quad (b \geq 1).$$

The group

$$C(t) = \{g \in \mathrm{GL}(V) : gt = tg\}$$

induces the full stabilizer of this filtration on $\ker t$. It also induces the full stabilizer of the corresponding filtration on $\ker t^ \subseteq V^*$, under the contragredient action. The algebraic group $C(t)$ is irreducible.*

Proof. Write V as a sum of Jordan row modules $\mathbb{C}[t]/(t^a)$. A module map from a row of length a to a row of length b can send its bottom nontrivially to the target bottom exactly when $a \leq b$. In that case, sending its top to a scalar multiple of t^{b-a} times the target top gives any desired bottom coefficient. These are precisely the matrix entries permitted by the filtration $F_b(t)$. Every filtration-preserving bottom matrix therefore lifts to a commuting endomorphism. If its action on $\ker t$ is invertible, the lift is invertible: a nonzero t -stable kernel would contain a nonzero vector in $\ker t$. This proves the first assertion. Apply the same argument to t^* and use inverse transpose for the dual assertion.

Finally, $C(t)$ is the determinant-nonzero open subset of the vector space $\text{End}_{\mathbb{C}[t]}(V)$, so it is irreducible. $\square$

6.4 The doubling construction

The threshold inequalities control every filtration level at which a new leaf bound can fail. After doubling they read $2K_r^T(b) - q_r(b) \leq 2L_r$ and $2K_r^S(b) - q_r(b) \leq 2M_r$. Lemma 6.4 will convert these inequalities into top and socle bounds at both new leaves, once their spaces are chosen with equal filtration profiles.

The doubled occurrence in Proposition 6.1 is also accessible through existing folding and stability results. Hong–Shen’s comparison for the fork automorphism carries occurrence for invariant D_n labels to saturated-cone membership for the corresponding C_{n-1} labels [15, Proposition 1.4(3)]. Zero-padding stability of the saturated cone in type C [11, Theorem 1.1], followed by Sam’s factor-two theorem [33, Theorem 1.1] and the reverse occurrence comparison [15, Proposition 1.4(1)], gives the doubled occurrence. At the bottom ranks one uses $D_3 = A_3$ and $D_2 = A_1 \times A_1$. We instead give the explicit module construction, which realizes the reduction while retaining the two sets of bounds at the new leaves.

Proof of Proposition 6.1. Occurrence implies that the simple-root coefficients of $\lambda + \mu - \xi$ are nonnegative integers, so choose a module from Proposition 6.2. Keep the chain through V_r and take two copies of it. The doubled space $\widetilde{W} = V_r \oplus V_r$ will be the new combined leaves. Write $\tilde{t} = t_r \oplus t_r$, $\tilde{u} = u_r \oplus u_r$, and $\tilde{v} = v_r \oplus v_r$.

Color every Jordan row of t_r alternately, and give its duplicate the opposite colors. Thus $\tilde{t}$ is off diagonal for the new splitting $\widetilde{W} = \widetilde{W}_+ \oplus \widetilde{W}_-$. Each color has dimension $\dim V_r$. A row of length a and its duplicate contribute one bottom vector of each color to every primal high-socle level with $b \leq a$; the dual vectors at their tops give the same contribution on the dual level. Thus both colors have dimension $q_r(b)$ on every primal and dual high-socle level, inside a level of total dimension $2q_r(b)$. Conjugating the coloring by $g \in C(\tilde{t})$ preserves all these properties.

We choose g to ensure the four leaf bounds. The subspace

$$A = \text{Ann}(\text{im } \tilde{u})$$

lies in $\ker \tilde{t}^*$, because $\text{im } \tilde{t} \subseteq \text{im } \tilde{u}$. Its dual filtration profile is $2K_r^T(b)$. Each dual color has profile $q_r(b)$. By Lemmas 6.4 and 6.5, a generic coloring has, in either dual color, intersection with A of dimension

$$\max \left(0, \max_b \{ 2K_r^T(b) - q_r(b) \} \right) \leq 2L_r,$$

where the last inequality is Lemma 6.3. This is exactly the new leaf top: a functional on a color annihilates the projected incoming image if and only if its extension by zero to the other color annihilates $\text{im } \tilde{u}$.

Similarly, $\ker \tilde{v} \subseteq \ker \tilde{t}$ has profile $2K_r^S(b)$. A generic primal color meets it in dimension at most $2M_r$, giving the new leaf socle bound. Each of these four requirements holds on a nonempty open subset of $C(\tilde{t})$. Since this group is irreducible, all four hold simultaneously. Surjectivity onto each of the primal and dual filtration stabilizers suffices here; the two actions need not be independently prescribable.

Let $P_\pm$ be the resulting color projections, and define the new leaf arrows by

$$u_\pm = P_\pm \tilde{u}, \quad v_\pm = \tilde{v}|_{\tilde{W}_\pm}.$$

Their return maps are zero because $P_\pm \tilde{u} \tilde{v}|_{\tilde{W}_\pm} = 0$. At the preceding trunk vertex,

$$v_+ u_+ + v_- u_- = \tilde{v}(P_+ + P_-) \tilde{u} = \tilde{v} \tilde{u}.$$

The relation, incoming-image sum, and outgoing kernel at that vertex are therefore unchanged from the doubled chain. All other trunk bounds simply double.

It remains to check the dimension vector. For $i < r$, the new trunk dimension is $2 \sum_{j \leq i} \Delta_j$. The two leaf coefficients for the D_{r+1} -weight

$$(2\Delta_1, \dots, 2\Delta_r, 0)$$

are both $\sum_{j \leq r} \Delta_j = \dim V_r$, precisely the two new color dimensions. The new leaf top bounds are $2L_r$, and the socle bounds are $2M_r$, as required. Applying Proposition 6.2 gives the asserted occurrence.

When $r = 1$, the retained edge is the virtual edge from $V_0 = 0$. Then $t_1 = u_1 = v_1 = 0$, and the construction produces the two disconnected vertex spaces of $D_2 = A_1 \times A_1$. The same bounds and dimension calculation apply. $\square$

6.5 Completion of the saturation theorem

Figure 2 records the changes of rank and scale in the common-zero-tail branch of the proof. In particular, the smaller-rank shape is obtained by restricting the original fundamental columns; it is not required to be a smaller-rank fundamental-column shape.

Proof of Theorem 1.1. Suppose first that the invariant at scale $N \in \mathbb{Z}_{>0}$ is nonzero. Set $\xi = \nu^*$, so $V(N\xi)$ occurs in $V(N\lambda) \otimes V(N\mu)$, and $\lambda + \mu - \xi$ lies in the root lattice. We may permute the three invariant labels before designating the moving label μ .

If one of them has at most one zero coordinate, choose it as μ . Lemma 2.5, followed by division by N , gives a real moment solution for λ, μ, ξ . Choose the shape η made of successive fundamental columns summing to μ . Theorem 3.1 gives an integral schedule for this shape, and Proposition 5.1 gives $V(\lambda) \otimes V(\mu) \longrightarrow V(\xi)$ nontrivially. This is the desired invariant.

Otherwise all three labels have at least two zero coordinates. They are then integer-coordinate weights, nonnegative and supported on initial coordinate segments. Choose a label of largest support as μ , and denote its support length by $r \leq n - 2$. If $r = 0$, all three representations are trivial. Assume $r \geq 1$. Apply Proposition 6.1 to the occurrence at scale N . It gives the occurrence in D_{r+1} for the truncated labels at scale $2N$. Lemma 2.5, divided by $2N$, therefore gives a real moment solution for the *unscaled* truncated labels.

Those labels remain integral, and their difference remains in $Q(D_{r+1})$: deleting zero coordinates preserves the even coordinate sum. The moving truncated label is positive in its first r coordinates and has exactly one zero. Take the fundamental column shape for the original D_n -weight μ . Since

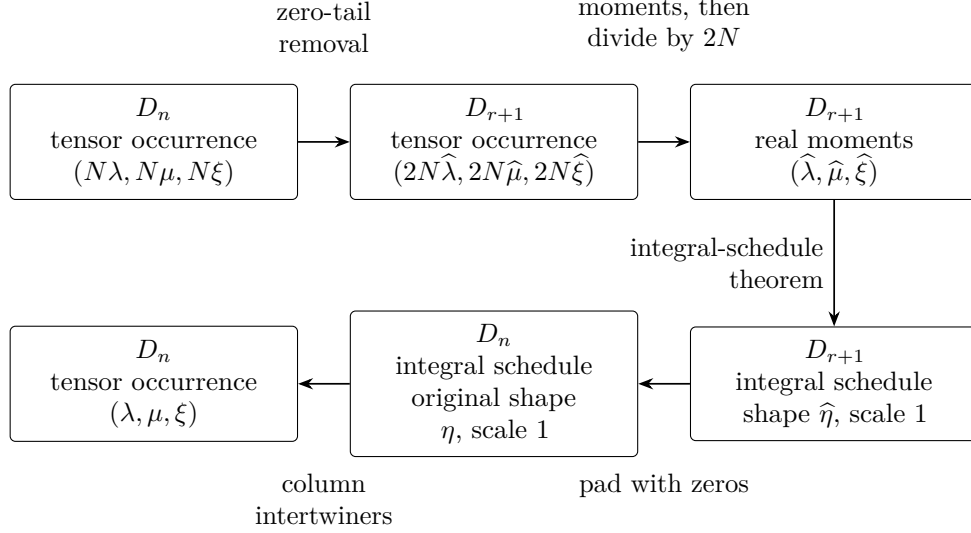


Figure 2: The common-zero-tail branch, with $1 \leq r \leq n - 2$. Here η is a fundamental-column shape for the original D_n moving weight, and $\hat{\eta}$ is its restriction to the first $r + 1$ coordinates. Doubling occurs only before passage to real moments. Both integral schedules and the final tensor occurrence use the original scale.

$M_{r+1} = \cdots = M_n = 0$, its fundamental expansion uses only columns $e_1 + \cdots + e_k$ with $k \leq r$; every increment therefore already vanishes beyond the retained coordinates. Restrict each increment to the first $r + 1$ coordinates. These increments are dominant for D_{r+1} . They need not be its fundamental weights; Theorem 3.1 applies to this arbitrary dominant-increment shape and gives an unscaled integral schedule.

Pad this schedule with zero coordinates. Its roots and even signed permutations belong to D_n , its endpoint lies in the required full Weyl orbit, and every switch coefficient and sign is unchanged. It is therefore an integral schedule for the *original* fundamental column shape. Proposition 5.1 again gives the unscaled invariant. The use of $2N$ supplied only real existence in smaller rank and has introduced no saturation factor.

Conversely, let an unscaled invariant functional be nonzero. The compact translates of highest-weight vectors span each irreducible representation, so the functional is nonzero on some triple of flag vectors. Its N -th tensor power is nonzero on the N -th powers of those same flag vectors. Those powers lie in the Cartan summands of highest weights $N\lambda, N\mu, N\nu$. Restriction gives the invariant at scale N .

The arguments include $D_2 = A_1 \times A_1$ and $D_3 = A_3$ with the stated conventions. If $\text{Spin}(2)$ is included, it is a torus; its representations are characters, and the assertion is the equivalence of the weight-sum equation with its positive integral multiple. This completes the proof. $\square$

Remark 6.6 (The stretching parameter). Positive integral stretching is essential to the formulation. If arbitrary positive rational N are allowed whenever all the scaled weights exist, the unscaled root-lattice condition alone does not suffice. For D_n , $n \geq 4$, take $\lambda = \mu = \nu = 2e_1$. The triangle contraction

$$(x, y)(y, z)(z, x)$$

of the standard orthogonal form defines a nonzero invariant on $V(2e_1)^{\otimes 3}$. Indeed isotropic squares lie in the highest-weight module $V(2e_1)$, and, in a hyperbolic basis a_i, b_i with $(a_i, b_j) = \delta_{ij}$, the

isotropic vectors

$$a_1, \quad b_1, \quad a_1 + b_1 + a_2 - b_2$$

have nonzero pairwise products. At $N = 1/2$ the three labels are e_1 , whose sum $3e_1$ is outside $Q(D_n)$; hence the triple has no invariant. This does not affect Theorem 1.1, where N is a positive integer.

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